Reduced matrix sequences and Gaussian cubic estimates for the real Grothendieck constant

October 2, 2026

Abstract

We construct an explicit sequence of rational matrices whose Grothendieck ratios converge to the reciprocal of a dimension-uniform Gaussian operator norm. Independently checked whole-line inequalities and a complete continuous source partition prove

$$\sup_d \|P_1 - (711/1000)P_3\|_{L^\infty(\gamma_d)\to L^1(\gamma_d)} \le \frac{40}{71}.$$

The resulting reduced matrix sequence proves $G_{\mathbb{R}} \ge 71/40 = 1.775$. A self-contained signed-tensor rounding argument gives $G_{\mathbb{R}} < 1760/987 < 1.785$, determining $G_{\mathbb{R}}$ to three significant digits as 1.78. Exact ratio-preserving maps impose symmetry, zero diagonal, zero row and column sums, and doubly stochastic entrywise absolute values. The analytic cell entries require $O(d)$ arithmetic operations after one-dimensional Gaussian moments are known. Fully rational finite-matrix lower bounds converge to 1.775 at rate $O(d^{-1})$; the actual matrix ratio limit is the reciprocal of the true operator norm. An exhaustive reduced sequence has monotone ratios converging to the full constant. The norm proof uses coupled cubic-chaos moment inequalities, scalar Stein certificates, and exact source geometry, with every numerical allowance bound to its complete local proof. The final lower bound does not use the unreconstructed historical coefficient-0.699 premise.

1 Introduction and the numerical target

For a nonzero finite real matrix A, define the bilinear norms

$$C(A) = \max_{s_i,t_j\in\{-1,1\}} \sum_{ij} A_{ij}s_it_j, \qquad V(A) = \max_{\|u_i\|=\|v_j\|=1} \sum_{ij} A_{ij}\langle u_i, v_j\rangle.$$

The vector dimension is arbitrary; finite dimension $m + n$ suffices for an m-by-n matrix. Write $G(A) = V(A)/C(A)$ and $G_{\mathbb{R}} = \sup_A G(A)$.

The complete norm certificate in Appendix L proves

$$\mathcal{N}(711/1000) := \sup_{d\ge1} \|P_1 - (711/1000)P_3\|_{L^\infty(\gamma_d)\to L^1(\gamma_d)} \le \frac{40}{71}, \qquad G_{\mathbb{R}} \ge \frac{71}{40} = 1.775. \tag{1}$$

Peng [1] proves $G_{\mathbb{R}} \le 17813/10000 = 1.7813$. Thus the resulting enclosure is

$$1.775 \le G_{\mathbb{R}} \le 1.7813,$$

which determines the constant to three significant digits as 1.78. Proposition 44 also proves the self-contained upper bound $G_{\mathbb{R}} < 1760/987 < 1.785$, sufficient for this rounding conclusion. The result is a three-significant-digit approximation, rather than a determination to three decimal places.

The explicit Gaussian–Hermite matrix family has actual ratio limit $1/\mathcal{N}(711/1000) \geq 1.775$. Its scalar norm is not identified with the certified cap, and this family is not assumed to attain the unrestricted constant. Proposition 1 separately gives an exhaustive reduced sequence whose ratios increase to $G_{\mathbb{R}}$. The logarithmic-cutoff construction and Corollary 47 give finite rational lower bounds and justify the limiting lower endpoint without numerical extrapolation. Proposition 46 expresses every analytic seed entry through one-dimensional truncated Gaussian moments.

The norm proof combines the coupled cumulant theorem in Section 5, the uniform cubic absolute-moment theorem in Section 6, the support–source comparisons in Section 7, and the actual cubic projection in Section 10. All local remainder allowances, target-coefficient spectral gates, and continuous source domains are checked independently. The final closed moment refinement keeps every earlier valid cap and exhausts all of its parent domains. The exact reduced-class maps are proved in Section 2. At $b = 1/2$, the published sharp operator inequality of Saha et al. [2] gives the exactly evaluable benchmark $6\pi/11$ in Corollary 2.

For historical comparison, the September 30 project estimate is retained solely as the separate all-dimension premise

$$\mathcal{N}(699/1000) \leq \frac{1000}{1773}. \tag{2}$$

The earlier report attributes this estimate to a 55-interval amplitude cover and inherited integrations [3]. That complete cover has not been reconstructed here. Assuming (2) gives $G_{\mathbb{R}} \geq 1.773$; the fresh coefficient-0.711 theorem and its finite matrix bounds have their own complete proof and do not use this premise.

2 The exact PAQ reduction

The right transformation is denoted R here to distinguish it from a Gaussian cubic Q. Suppose distinct columns of P have disjoint supports, and distinct rows of R have disjoint supports. Put

$$p_i = \sum_a |P_{ai}|, \qquad r_j = \sum_b |R_{jb}|.$$

Then

$$C(PAR) = C(D_p A D_r), \qquad V(PAR) = V(D_p A D_r). \tag{3}$$

In particular, if $p_i = \alpha > 0$ for every original row and $r_j = \beta > 0$ for every original column, both norms are multiplied by $\alpha\beta$, and $G(PAR) = G(A)$.

For scalar signs, the aggregate in row group i lies in $[-p_i, p_i]$. The maximum of the resulting bilinear form on the product of these intervals occurs at a vertex, and every vertex is attainable by aligned copied signs. For vectors the aggregate lies in the ball of radius p_i; aligned copies attain its boundary. Ball vectors may be completed to unit vectors using separate left and right orthogonal coordinates without changing cross inner products. This proves (3) for both norms.

Signed permutations and common scaling are special cases. Ordinary diagonal balancing and arbitrary orthogonal changes of coordinates are not consequences of this theorem. The requirements concern total absolute copying weights, not the number of copies.

2.1 A universal absolute doubly stochastic class

Let Z be a nonzero integer matrix with zero rows and columns removed. Set

$$r_i = \sum_j |Z_{ij}|, \qquad c_j = \sum_i |Z_{ij}|, \qquad L = \sum_i r_i = \sum_j c_j.$$

Make r_i copies of row i, and c_j copies of column j, and define the square matrix of order L by

$$\boxed{B_{(i,a),(j,b)} = \frac{Z_{ij}}{r_i c_j}, \quad 1 \le a \le r_i, \quad 1 \le b \le c_j.} \tag{4}$$

Each original row or column has total copying weight one. Thus

$$C(B) = C(Z), \qquad V(B) = V(Z), \qquad G(B) = G(Z). \tag{5}$$

Every row and column of $|B|$ sums to one. For example, a cloned row has sum $\sum_j c_j |Z_{ij}|/(r_i c_j) = 1$.

For rational input first clear its denominator. For real input choose rational approximations; both norms satisfy

$$|C(A) - C(A')|, \ |V(A) - V(A')| \le \sum_{ij} |A_{ij} - A'_{ij}|. \tag{6}$$

Consequently the supremum defining $G_{\mathbb{R}}$ is unchanged by restricting to rational matrices whose entrywise absolute values are doubly stochastic, over all orders. The clone order can be enormous, but both objectives can be evaluated on the smaller seed Z.

2.2 Symmetry, zero diagonal and zero sums

For B in (4), set

$$K(B) = \begin{pmatrix} 0 & B \\ B^T & 0 \end{pmatrix}, \qquad J_0 = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad H(B) = J_0 \otimes K(B). \tag{7}$$

Then $H(B)$ is rational, symmetric, zero-diagonal, has zero row and column sums, and $|H(B)|$ is doubly stochastic. Its order is $4L$ and

$$C(H(B)) = 4C(B), \qquad V(H(B)) = 4V(B). \tag{8}$$

The two off-diagonal terms in $K(B)$ use independent left/right variable lists and can optimize simultaneously, giving factor two. The signed two-copy map J_0 gives a further factor two by (3). Hence all these restrictions can be imposed simultaneously without changing the supremum or any individual rational seed ratio.

2.3 A monotone sequence converging to the full constant

The analytic cubic sequence below has limit $1/\mathcal{N}(b)$. The reduction also permits an unconditional sequence converging to the full Grothendieck constant, without assuming that a cubic operator is optimal.

Proposition 1. *Let $\mathcal{A}_n$ consist of all nonzero real matrices of dimensions at most n whose entries lie in $2^{-n}\mathbb{Z} \cap [-1, 1]$. Choose a matrix $S_n \in \mathcal{A}_n$ maximizing G, with a fixed lexicographic rule for ties. Clear its dyadic denominator and apply* (4) *and* (7)*, obtaining $\widetilde{H}_n$. Then every $\widetilde{H}_n$ belongs to the simultaneous reduced class and*

$$G(\widetilde{H}_n) \le G(\widetilde{H}_{n+1}), \qquad \lim_{n\to\infty} G(\widetilde{H}_n) = G_{\mathbb{R}}.$$

Proof. Each candidate set is finite and the sets are nested. Both norms are positive on a nonzero matrix, and the exact reductions preserve their ratio, so the maxima exist and increase monotonically. For any fixed nonzero real matrix A, homogeneity permits the normalization $\max_{ij} |A_{ij}| \le 1$. Its dyadic approximants $A^{(n)}$ belong to $\mathcal{A}_n$ for sufficiently large n and have total entry error tending to zero. The two norms satisfy

$$|C(A) - C(B)| \le \sum_{ij} |A_{ij} - B_{ij}|, \qquad |V(A) - V(B)| \le \sum_{ij} |A_{ij} - B_{ij}|.$$

Consequently $G(A^{(n)}) \to G(A)$ and $\liminf_n G(\widetilde{H}_n) \ge G(A)$. Taking the supremum over A proves the asserted limit; the reverse inequality follows from the definition of $G_{\mathbb{R}}$. □

This is a finite exact specification at every stage. The scalar norm is computed by sign enumeration. The vector norm is the maximum of the rational linear functional $\sum_{ij} A_{ij} X_{i,m+j}$ on the compact set of positive semidefinite $(m+n)$-by-$(m+n)$ matrices with diagonal one. Its polynomial description makes exact algebraic optimization and comparison possible. Enumeration is computationally expensive. The analytic subfamily gives controlled candidates and a quantitative limit, while retaining a best previous candidate prevents a weaker trial from replacing it.

3 An explicit analytic seed and its limit

Let $b \in [0, 1)$ be computable (a rational coefficient suffices), and put $T_b = P_1 - bP_3$, where P_j projects onto total Hermite degree j in Gaussian space. Start at $n \ge 2$ and let $d_n = n^2$. In each coordinate partition the line by the grid of mesh n^{-2} from $-n$ to n, together with its two exterior intervals. The grid has zero as an edge and is antipodally symmetric. Form product cells and retain the cells whose first coordinate lies on the negative half-line. These represent all antipodal pairs.

For a normalized product Hermite H_ξ, define

$$F_{j,n}[C, \xi] = 2 \int_C H_\xi \, d\gamma_{d_n}, \qquad |\xi| = j, \quad j \in \{1, 3\}.$$

The finite seed is

$$A_n(b) = F_{1,n} F_{1,n}^T - b F_{3,n} F_{3,n}^T. \tag{9}$$

All entries are analytic and computable. With normalized one-variable Hermites $h_k = \mathrm{He}_k / \sqrt{k!}$,

$$\int_a^b h_0 \, d\gamma = \Phi(b) - \Phi(a), \qquad \int_a^b h_k \, d\gamma = \frac{\varphi(a)\mathrm{He}_{k-1}(a) - \varphi(b)\mathrm{He}_{k-1}(b)}{\sqrt{k!}} \quad (k \ge 1),$$

with boundary terms zero at infinite endpoints. Products give the multivariate entries, so there is no unevaluated high-dimensional integral.

Extending cell signs oddly shows $C(A_n(b)) \le \mathcal{N}(b)$, where $\mathcal{N}(b)$ is the supremum of the scalar operator norm over all Gaussian dimensions. Conversely any bounded odd functions in a fixed dimension can be approximated by these partitions in their first coordinates as $n \to \infty$. Conditional cell averaging converges in L_2; the remaining coordinates integrate out. The partitions need not be nested: continuous compactly supported functions are dense in L_2, and their cell-average error tends to zero as the mesh shrinks and the cutoff grows. Conditional averaging is an L_2 contraction, so density gives the same conclusion for arbitrary L_2 functions. Therefore

$$C(A_n(b)) \longrightarrow \mathcal{N}(b). \tag{10}$$

The supremum can be restricted to odd functions because T_b annihilates the even part. Maximization over bounded step functions reduces to signs at the vertices of the finite cubes.

For $X \sim \gamma_d$, the radial vector field $X/\|X\|$ gives

$$q_d(b) = a_d \left(1 - \frac{b}{2(d+2)}\right), \qquad a_d = \frac{2\Gamma((d+1)/2)^2}{d\Gamma(d/2)^2} \longrightarrow 1. \tag{11}$$

Its total degree-one energy is a_d; its degree-three energy is $a_d/[2(d+2)]$, by Gaussian radial integration. The ordinary L_2 norm of T_b is one. Consequently $V(A_n(b)) \le 1$.

For an explicit approximation bound, let $r_n(X)$ be the vector of cell midpoints, with exterior coordinates clipped to $\pm n$, and set $v_n = r_n/\|r_n\|$. All its coordinates are nonzero because zero is an edge of the grid. Thus this is an odd unit vector field, constant on cells. With $d = n^2$, $h = n^{-2}$, and

$$p_n = \min\{1, 2n\varphi(n)\}, \qquad E_n = \sqrt{\frac{1}{n^2(n^2-2)} + 8n\varphi(n)},$$

the elementary normalization inequality and Gaussian Mills bound give

$$\|v_n - X/\|X\|\|_{L_2}^2 \le dh^2 \mathbb{E}\|X\|^{-2} + 4\Pr(\max_i |X_i| > n) \le E_n^2. \tag{12}$$

Here $\mathbb{E}\|X\|^{-2} = 1/(d-2)$, and the union bound gives $\Pr(\max_i |X_i| > n) \le 2d\varphi(n)/n$. In particular $E_n = n^{-2} + O(n^{-4})$.

The vector objective admits a sharper, quadratic error estimate. Write $R = \|X\|$, $\mu = \mathbb{E}R$, $w = X/R$, and $e = v_n - w$. The degree-one and degree-three projections above give

$$T_b w = t(R)w, \qquad t(R) = \frac{\mu R}{d}\left(1 - \frac{b}{2} + \frac{bR^2}{2(d+2)}\right) \ge 0.$$

Pointwise unit length implies $e \cdot w = -\|e\|^2/2$. Since the lowest odd spectral coefficient of T_b is $-b$,

$$\langle w, T_b w\rangle - \langle v_n, T_b v_n\rangle = \mathbb{E}[t(R)\|e\|^2] - \langle e, T_b e\rangle \le \mathbb{E}[(t(R) + b)\|e\|^2].$$

Off the exterior event, $\|e\|^2 \le dh^2/R^2$; on it, $\|e\|^2 \le 4$. Using $\mu\mathbb{E}R^{-1} = da_d/(d-1)$, $a_d \le 1$, and

$$\mathbb{E}[R\mathbf{1}_E] \le \sqrt{d\Pr(E)}, \qquad \mathbb{E}[R^3\mathbf{1}_E] \le \sqrt{d(d+2)(d+4)\Pr(E)},$$

proves the entirely explicit bound

$$\begin{aligned} L_n(b) = \frac{1}{n^2}&\left[\frac{1 - b/2}{n^2 - 1} + \frac{b}{2(n^2+2)} + \frac{b}{n^2-2}\right] \\ &+ 4\left[1 - \frac{b}{2} + \frac{b}{2}\sqrt{\frac{n^2+4}{n^2+2}}\right]\sqrt{p_n} + 4bp_n. \end{aligned} \tag{13}$$

Thus

$$V(A_n(b)) \ge q_{n^2}(b) - L_n(b) \longrightarrow 1, \qquad L_n(b) = (1+b)n^{-4} + O(n^{-6}). \tag{14}$$

The exponentially small terms in (13) are retained in every finite bound. The first-order direction error cancels because the radial field is pointwise aligned with its operator image.

3.1 The sequence inside the reduced class

Choose computable symmetric dyadic matrices $\widehat{A}_n(b)$ with total entry error $\eta_n \le n^{-4}$. Directed entry intervals select the first dyadic approximation whose certified error meets that budget; this avoids deciding an exact rounding tie. Clear the denominator to $Z_n(b)$, apply (4), and then (7):

$$\boxed{H_n(b) = H(B(Z_n(b))).} \tag{15}$$

Every $H_n(b)$ lies in the simultaneous reduced class, and

$$G(H_n(b)) = G(\widehat{A}_n(b)) \ge \frac{q_{n^2}(b) - L_n(b) - \eta_n}{\mathcal{N}(b) + \eta_n}.$$

Indeed $C(A_n(b))$ is bounded away from zero: a first-coordinate hemisphere gives $(2/\pi)(1 - b/6) > 5/(3\pi) > 1/4$. Thus $n \ge 2$ also ensures $\widehat{A}_n(b) \ne 0$. Equations (6)–(14) prove

$$\lim_{n\to\infty} G(H_n(b)) = \frac{1}{\mathcal{N}(b)}. \tag{16}$$

In particular (2) proves $\lim_n G(H_n(.699)) \ge 1.773$ assuming that premise. The exact normalization preserves this endpoint. The gamma recurrence gives $a_{d+2}/a_d = (d+1)^2/[d(d+2)]$, and consequently $a_d > 1 - 1/(2d)$: along each parity the quotient $a_d/(1 - 1/(2d))$ decreases to one. Therefore any proved $\mathcal{N}(b) \le B$ gives the finite, computable lower bound

$$G(H_n(b)) \ge \frac{(1 - 1/(2n^2))(1 - b/[2(n^2+2)]) - L_n(b) - n^{-4}}{B + n^{-4}} = \frac{1}{B} - \frac{1+b}{2Bn^2} + O(n^{-4}). \tag{17}$$

Corollary 2 (An exactly evaluable Gaussian benchmark)**.** *Let $T = P_1 - \frac{1}{2}P_3$ on standard Gaussian space and let A_n be its antipodally folded product-cell compression, with dimension $d = n^2$, mesh n^{-2}, cutoff $\pm n$, and 0 a coordinate-cell boundary. Then, for every $n \ge 2$,*

$$C(A_n) = \frac{11}{6\pi}, \qquad V(A_n) \le 1, \qquad \lim_{n\to\infty} G(A_n) = \frac{6\pi}{11}.$$

Choose a symmetric rational approximation $\widehat{A}_n$ with total entrywise error at most $\eta_n = n^{-4}$, and apply the exact cloning, symmetric dilation and balancing reductions to obtain H_n. Thus H_n is symmetric, has zero diagonal and zero row sums, and $|H_n|$ is doubly stochastic. The same actual ratio limit holds:

$$\lim_{n\to\infty} G(H_n) = \frac{6\pi}{11} = 1.713595992867159948\ldots.$$

The rationalized ratios need not lie below their limiting value.

Proof. The all-odd-sign inequality proved by Saha et al. [2], Sections 8–9, is

$$2\langle P_1 f, P_1 g\rangle - \langle P_3 f, P_3 g\rangle \le \frac{11}{6}\frac{2}{\pi}.$$

Replacing g by $-g$ gives the corresponding absolute bound. It also holds for odd functions of absolute value at most one: successively maximize the bilinear pairing in each function, choosing the odd sign of the other function's image under T. For $s(X) = \operatorname{sgn}(X_1)$ and $\nu = \sqrt{2/\pi}$, the raw Hermite polynomial $\mathrm{He}_3(z) = z^3 - 3z$ gives

$$P_1 s = \nu X_1, \qquad P_3 s = -\frac{\nu}{6}\,\mathrm{He}_3(X_1), \qquad \langle s, Ts\rangle = \nu^2 - \frac{\nu^2}{12} = \frac{11}{6\pi}.$$

The hemisphere sign is exactly constant on each product cell. Consequently its folded matrix assignment attains the external upper bound, establishing the exact scalar norm for every finite n. For any pair of unit-vector assignments, their odd cell fields v, w have $\int \|v\|^2 \, d\gamma_d = \int \|w\|^2 \, d\gamma_d = 1$; the L^2 operator norm $\|T\|_{2\to 2} = 1$ and Cauchy–Schwarz give $V(A_n) \le 1$. The radial unit field and the established midpoint estimate give $V(A_n) \to 1$. Finally, total entrywise error bounds both norm errors by η_n, and every subsequent reduction preserves G exactly. This proves both asserted limits. □

For completeness, the following are explicit finite lower bounds. Put $d = n^2$, $p_n = 2n\varphi(n)$, and

$$\begin{aligned} e_n &= \frac{1}{d}\left(\frac{3}{4(d-1)} + \frac{1}{4(d+2)} + \frac{1}{2(d-2)}\right) \\ &\quad + 4\left(\frac{3}{4} + \frac{1}{4}\sqrt{\frac{d+4}{d+2}}\right)\sqrt{p_n} + 2p_n, \\ q_n &= \left(1 - \frac{1}{2d}\right)\left(1 - \frac{1}{4(d+2)}\right) - e_n. \end{aligned}$$

The gamma-ratio lower bound $a_d > 1 - 1/(2d)$ and the quadratic midpoint loss imply

$$G(A_n) \ge \frac{q_n}{11/(6\pi)}, \qquad G(H_n) \ge \frac{q_n - n^{-4}}{11/(6\pi) + n^{-4}}.$$

Directed MPFR256 evaluation, rounded down to twelve decimal places, gives

n	$G(A_n)$ lower bound	$G(H_n)$ lower bound
32	1.712339483749	1.712335051217
128	1.713517545081	1.713517527758
1024	1.713594767206	1.713594767202

These evaluations are recorded in `saha_benchmark_certificate.json`; the external scalar theorem is the sole new norm input and has not been independently replayed here. This benchmark does not invoke the retained, unverified global norm premise used for the earlier 1.773 candidate.

The same external theorem determines the complete analytic range $0 \le b \le 1/2$. Indeed $T_b = 2bT_{1/2} + (1-2b)P_1$, and $\mathcal{N}(0) = 2/\pi$. Convexity and the same hemisphere equality give

$$\mathcal{N}(b) = \frac{2}{\pi}(1 - b/6), \qquad \lim_n G(H_n(b)) = \frac{\pi}{2(1-b/6)} \quad (0 \le b \le 1/2).$$

This exactly soluble range ends at $6\pi/11$; improving it requires $b > 1/2$ or a different operator family.

4 Coefficient choices and construction ceilings

The final certified sequence uses $b = 711/1000$, with the complete norm cap $40/71$ in Theorem 48, proving the lower endpoint $71/40 = 1.775$. The following construction ceilings describe earlier coefficient choices and their reflected sign witnesses.

The initial candidate uses $b = 18/25 = .72$ in (9)–(15). For this coefficient, the target requires the all-dimension inequality

$$\boxed{\mathcal{N}(18/25) \le \frac{40}{71}.} \qquad \frac{1}{40/71} = 1.775. \tag{18}$$

The previous .699 operator cannot reach this target: its retained directed diffuse-cubic sign witness limits its reciprocal-norm construction to $1.77440101857\ldots$ [5]. Changing the coefficient is therefore necessary within the pure cubic family.

The .72 coefficient has a fresh directed lower witness. Let $h_3(x) = (x^3-3x)/\sqrt{6}$, let all Gaussians below be independent, and put

$$Y_N = N^{-1/2}\sum_{j=1}^{N} h_3(X_j), \qquad a = 511/200, \quad c = 393/1000,$$
$$f_N = \operatorname{sgn}(Y_N + aZ + cZ^3), \qquad g_N = \operatorname{sgn}(-Y_N + aZ + cZ^3).$$

The spectral central limit calculation assigns ordinary Z-degree j and diffuse-cubic degree k to total original Hermite degree $j + 3k$. Its full degree accounting is proved in Appendix E. Writing $q(z) = az + cz^3$, the limiting moments are

$$m_Y = 2\int_{\mathbb{R}} \varphi(z)\varphi(q(z))\,dz, \quad m_1 = 2\int_{\mathbb{R}} \varphi(z)q'(z)\varphi(q(z))\,dz,$$
$$m_3 = \frac{2}{\sqrt{6}}\int_{\mathbb{R}} (z^2-1)\varphi(z)q'(z)\varphi(q(z))\,dz.$$

The scalar pairing is $m_1^2 - b(m_3^2 - m_Y^2)$. Directed Taylor integration and explicit tails give

$$\mathcal{N}(.72) > .563241882518393914882, \qquad \frac{1}{\mathcal{N}(.72)} < 1.7754361511. \tag{19}$$

This is a ceiling for the construction, not an achieved lower bound. The witness lies below $40/71$ by more than .00013839917, and hence does not rule out (18). Old whole-frame profile transfers fail to certify (18); their failure is a limitation of those sufficient estimates, not an obstruction from the true norm.

Proposition 3 (Exact reflected norm witness at $b = 7171/10000$)**.** *Let* $\mathcal{N}(b) = \sup_d \|P_1 - bP_3\|_{L^\infty(\gamma_d)\to L^1(\gamma_d)}$. *Then*

$$\mathcal{N}\left(\frac{7171}{10000}\right) > \frac{56323352479769}{10^{14}}.$$

Consequently the asymptotically unit-numerator reciprocal-norm construction for this fixed coefficient has ceiling strictly below $1775462496412/10^{12} = 1.775462496412$. *This is a construction ceiling, not a Grothendieck lower endpoint.*

Proof. Use the normalized Hermite $h_3(x) = (x^3-3x)/\sqrt{6}$. Set $a = 26412631/10^7$, $c = 4084047/10^7$, $q(z) = az + cz^3$ and $Y_n = n^{-1/2}\sum_{j=1}^n h_3(X_j)$ for independent standard Gaussian variables. Use the actual original-space signs $f_n = \operatorname{sgn}(Y_n + q(Z))$ and $g_n = \operatorname{sgn}(-Y_n + q(Z))$. The original-degree projection accounting for the reflected sign witness applies because $q'(z) \geq a > 0$; in particular the diffuse channel retains original degree three. Thus the limiting first- and third-chaos pairings are m_1^2 and $m_3^2 - m_Y^2$, respectively, where

$$m_Y = \frac{2}{\pi}\int_0^\infty e^{-(z^2+q(z)^2)/2}\,dz, \quad m_1 = \frac{2}{\pi}\int_0^\infty q'(z)e^{-(z^2+q(z)^2)/2}\,dz,$$
$$m_3 = \frac{2}{\pi\sqrt{6}}\int_0^\infty (z^2-1)q'(z)e^{-(z^2+q(z)^2)/2}\,dz.$$

The exact-rational directed certificate `certify_cubic7171_norm_witness.py` integrates Taylor terms through degree eight on 320 rational cells of $[0,5]$, bounds the degree-ten remainder over each entire cell, and bounds the omitted Gaussian tails explicitly. At 384-bit MPFR precision it encloses

$$m_1^2 - \frac{7171}{10000}(m_3^2 - m_Y^2) \in [0.563233524797690133779 3,\ 0.563233524797690134435 6].$$

Every finite-dimensional sign pairing is bounded above by $\mathcal{N}(7171/10000)$; taking the limit proves the assertion. The reciprocal enclosure has an upper endpoint below 1.77546249641158, proving the stated construction ceiling. The witness remains below $40/71$ and therefore does not exclude a future certified lower endpoint of 1.775. A separate Arb replay at 512 bits, Taylor order twelve and 400 rational cells recomputed all three integrals and Gaussian tails without importing the producer, independently accepting the stated inequalities. □

A fresh reflected-sign witness also excludes $b = 141/200 = .705$ from reaching the target within this reciprocal-norm family. Use the exact head $q(z) = (31091537/10^7)z + (49105809/10^8)z^3$ in the same construction. Directed integration gives

$$\mathcal{N}(.705) > \frac{563383848679}{10^{12}} > \frac{40}{71}, \qquad \frac{1}{\mathcal{N}(.705)} < 1.774988761828 < 1.775.$$

An independent 512-bit Arb replay, with Taylor order twelve and 400 rational cells, checks all three moments and the omitted Gaussian tails. The same replay at $b = .713$ gives a construction ceiling below 1.775407850864, leaving the target possible at the base coefficient $b = .713$. These are ceilings for the specified matrix limits, and impose no upper bound on $G_{\mathbb{R}}$.

5 A coupled cumulant inequality for real cubic chaos

Let $Q = I_3(f)$ be a real third Gaussian chaos, normalized by $\mathbb{E}Q^2 = 6\|f\|_F^2 = 1$, where f is a real fully symmetric third-order tensor. The flattening Gram matrices below are symmetric matrices; they need not be fully symmetric as four-index tensors. The norm $\|\cdot\|_F$ is the Euclidean norm of all entries. Define

$$\kappa = \mathbb{E}Q^4 - 3, \qquad \kappa_6 = \mathbb{E}Q^6 - 15\mathbb{E}Q^4 + 30, \qquad U = \tfrac{1}{3}\|DQ\|^2 - 1, \qquad K = 194/\sqrt{10}.$$

Theorem 4. *In every finite Gaussian dimension,*

$$\kappa_6 \le K\kappa^{3/2} - 633\sqrt{\kappa}\left(\mathbb{E}U^2 - \frac{7\kappa}{45}\right). \tag{20}$$

Also $0 < \kappa \le 90$ *and* $7\kappa/45 \le \mathbb{E}U^2 \le 2\kappa/9$. *Equality holds in* (20) *for* $Q = (Z^3 - 3Z)/\sqrt{6}$.

The scalar endpoint $\kappa = 0$ is included by continuity in the later price calculations. A nonzero normalized cubic tensor in finite dimension has $\kappa > 0$.

Write $F_{ab,c} = f_{abc}$, $A = FF^T$, $G = F^TF$, and let S be the full symmetrization of the four-index tensor A. Put

$$s = \|A\|_F^2 = \|G\|_F^2, \quad h = \|S\|_F^2, \quad V = \operatorname{tr} A^3, \quad V_4 = \operatorname{tr} A^4.$$

Symmetrization is an orthogonal projection, so $\langle A, S\rangle = h$, $0 \le h \le s$, and

$$\|S + tA\|_F^2 = t^2 s + (1 + 2t)h \qquad (t \in \mathbb{R}). \tag{21}$$

The Gaussian product formula gives

$$U = 3I_4(S) + 12I_2(G), \quad \mathbb{E}U^2 = 216h + 288s, \quad \kappa = 1296s + 1944h. \tag{22}$$

For example, the first identity follows by applying $D_i I_3(f) = 3I_2(f_i)$ and

$$I_2(g)^2 = I_4(\mathrm{Sym}(g \otimes g)) + 4I_2(g \otimes_1 g) + 2\|g\|_F^2$$

to each tensor slice. Orthogonality of Hermite degrees gives the variance. Since $\operatorname{tr} A = 1/6$, $s \le 1/36$, which also proves $\kappa \le 90$.

The sixth-cumulant identity is

$$\frac{\kappa_6}{46656} = \frac{135}{2}\operatorname{tr} S^3 + 45\operatorname{tr}(AS^2) + \frac{165}{2}\operatorname{tr}(A^2S) - 20\operatorname{tr} A^3 + \frac{135}{2}\langle S, G \otimes G\rangle. \tag{23}$$

It agrees with the real cubic identity in [7, Eq. (8.12)]. For completeness, the accompanying exact program derives it from connected Wick diagrams. For r copies of f, a loopless multigraph with multiplicities m_{ij} and degree three at each vertex has weight $(3!)^r/\prod_{i<j} m_{ij}!$. Cumulants retain precisely the connected graphs, by the exponential formula for the logarithm of the Wick generating function. There are seven labelled connected graphs in the fourth cumulant and 640 in the sixth. Expanding each term of (23), including the three pairings in S, gives the same rational coefficient in each of the six sixth-order graph classes. This checks a formal tensor identity; finite examples of tensors are not substituted for that identity.

Set

$$y = \sqrt{h/s}, \qquad a = (V/s^{3/2})^{1/3}, \qquad d = 1 + \tfrac{3}{2}y^2.$$

Then $0 \le y, a \le 1$ and

$$\kappa = 1296sd, \qquad \frac{\mathbb{E}U^2}{\kappa} = \frac{1}{9}\left(1 + \frac{1}{d}\right). \tag{24}$$

The stated variance interval follows immediately from $1 \le d \le 5/2$.

For any of the four pairs

$$(p, q) = (17/33, -1),\ (62/105, -3/4),\ (28/39, -1/2),\ (3/4, -1),$$

put

$$c_p = \frac{55/2 - 15q}{p(p-q)}, \quad c_q = \frac{55/2 - 15p}{q(q-p)}, \quad \zeta = 135/2 - c_p - c_q, \quad \beta = 20 + c_p p^3 + c_q q^3.$$

These pairs have $c_p, c_q > 0$ and $\zeta \ge 0$. Direct expansion completes the first four terms on the right of (23) as

$$c_p \operatorname{tr}(S + pA)^3 + c_q \operatorname{tr}(S + qA)^3 + \zeta \operatorname{tr} S^3 - \beta V.$$

For a real symmetric matrix B, $\operatorname{tr} B^3 \le \|B\|_F^3$. Also

$$\|\mathrm{Sym}(G \otimes G)\|_F^2 = (s^2 + 2V_4)/3, \qquad V_4 \le V^{4/3}.$$

Together with (21), these inequalities prove $\kappa_6/\kappa^{3/2} \le \mathcal{J}_{p,q}(y, a)/d^{3/2}$, where

$$\begin{aligned}\mathcal{J}_{p,q} = {} & c_p[p^2 + (1+2p)y^2]^{3/2} + c_q[q^2 + (1+2q)y^2]^{3/2} \\ & + \zeta y^3 - \beta a^3 + \frac{135}{2} y\sqrt{(1+2a^4)/3}.\end{aligned} \tag{25}$$

An additional valid bound is $\mathcal{M}(y,a)/d^{3/2}$, with

$$\mathcal{M} = \frac{135}{2}y^3 + \frac{135}{2}y\sqrt{(1+2a^4)/3} + 45ay^2 - 20a^3 \\ + \frac{165}{2}\min\left\{a^{3/2}y, y^2a^3 + ya^2\sqrt{(1-y^2)(1-a^2)}\right\}. \tag{26}$$

For the first mixed bound use $\operatorname{tr}(A^2S) \le \sqrt{hV_4}$ and $V_4 \le \sqrt{s}V$. For the second, write $R = S-(h/s)A$ and apply Cauchy–Schwarz to $\langle A^2 - (V/s)A, R\rangle$, then use $V_4 \le V^{4/3}$. Finally $\operatorname{tr}(AS^2) \le V^{1/3}h$ since $\|A\|_{\mathrm{op}} \le V^{1/3}$. No positivity of S is assumed.

The exact scalar comparison, supplied in Appendix A, is

$$\frac{\min\{\mathcal{M}, \mathcal{J}_{17/33,-1}, \mathcal{J}_{62/105,-3/4}, \mathcal{J}_{28/39,-1/2}, \mathcal{J}_{3/4,-1}\}}{d^{3/2}} + \frac{211}{3d} \le \frac{194}{\sqrt{10}} + \frac{422}{15} \quad (0 \le y, a \le 1). \tag{27}$$

Its certificate consists of 84 rational rectangles and 2100 exact Bernstein coefficient checks in $\mathbb{Q}(\sqrt{10})$. Combining (24)–(27) with

$$633\left\{\frac{1}{9}(1+1/d) - \frac{7}{45}\right\} = \frac{211}{3d} - \frac{422}{15}$$

proves Theorem 4. At $Q = (Z^3-3Z)/\sqrt{6}$, $y = a = 1$, $\kappa = 90$, and $\kappa_6 = 52380 = K90^{3/2}$.

6 A uniform cubic absolute-moment bound

Theorem 5. *For every normalized real third Gaussian chaos Q in every finite dimension,*

$$\mathbb{E}|Q| < \rho := 849/1000, \qquad \|P_3\|^2_{L_\infty \to L_2} \le 720801/1000000. \tag{28}$$

Put $c = \sqrt{2/\pi}$ and $R(x) = e^{x^2/2}\int_x^\infty e^{-t^2/2}\,dt$ for $x \ge 0$. The odd extension of $g(x) = -1+cR(x)$ solves $g'(x) - xg(x) = |x| - c$, with

$$g'(x) = -c(1 - |x|R(|x|)).$$

The cubic Gaussian Stein identity is

$$\mathbb{E}[Qv(Q)] = \mathbb{E}[(1+U)v'(Q)], \qquad U = \tfrac{1}{3}\|DQ\|^2 - 1. \tag{29}$$

It follows directly from Gaussian integration by parts and the Ornstein–Uhlenbeck eigenvalue of Q. Consequently, for every odd continuously differentiable regression ψ of polynomial growth and every $\eta > 0$,

$$\mathbb{E}|Q| \le c + \mathbb{E}\left[\psi'(Q) - Q\psi(Q) + \frac{(g'(Q)-\psi'(Q))^2}{2\eta}\right] + \frac{\eta}{2}\mathbb{E}U^2. \tag{30}$$

Indeed $\mathbb{E}|Q| - c = -\mathbb{E}[Ug'(Q)]$ and $\mathbb{E}[\psi'(Q) - Q\psi(Q)] = -\mathbb{E}[U\psi'(Q)]$; Young's inequality applied to $-U(g'-\psi')$ gives (30).

The eight rational rows in Appendix B supply $\eta > 0$, coefficients q_0, q_2, q_4, q_6 with $q_6 > 0$, and a regression ψ, such that on the entire real line

$$\psi'(x) - x\psi(x) + \frac{(g'(x)-\psi'(x))^2}{2\eta} \le q_0 + q_2x^2 + q_4x^4 + q_6x^6. \tag{31}$$

For $x \geq 0$, each regression has the explicit rational form

$$\psi(x) = v_0 x + \frac{v_1 x^2}{2} + \frac{v_2 x^3}{3} + \frac{v_3 x^4}{4} + \frac{x}{x+2} \sum_{j=0}^{12} v_{4+j} T_j \left(\frac{x-2}{x+2} \right), \tag{32}$$

extended oddly. It is continuously differentiable at zero. All v_j, all q_j, and η are rational and are supplied in the source archive as exact inputs, separately from floating discovery output.

Combining (30)–(31) with Theorem 4 gives a price affine in $\mathbb{E}U^2$. Its maximum therefore occurs at one of the two permitted variance endpoints. The resulting functions of κ are

$$\begin{aligned} p_0(\kappa) &= c + q_0 + q_2 + (3+\kappa)q_4 + [15 + 15\kappa + K\kappa^{3/2}]q_6 + 7\eta\kappa/90, \\ p_1(\kappa) &= c + q_0 + q_2 + (3+\kappa)q_4 + [15 + 15\kappa + (K - 633/15)\kappa^{3/2}]q_6 + \eta\kappa/9. \end{aligned} \tag{33}$$

Both are convex for $\kappa \geq 0$ because $q_6 > 0$ and $K - 633/15 > 0$. On a closed cumulant interval, four corner prices thus suffice. The eight intervals in Appendix B cover $[0, 90]$, and every directed corner enclosure is below $.848521 < .849$. This proves the first assertion of Theorem 5.

For the second assertion, normalize $P_3 f$ to a variance-one cubic Q when that projection is nonzero. For $|f| \leq 1$, $\|P_3 f\|_2 = \mathbb{E} f Q \leq \mathbb{E}|Q| < \rho$. Taking the supremum and squaring proves the projection norm bound.

Theorem 6 (A refitted uniform cubic absolute-moment bound)**.** *For every normalized real third Gaussian chaos Q in every finite dimension,*

$$\mathbb{E}|Q| < \frac{847}{1000}, \qquad \|P_3\|^2_{L_\infty \to L_2} \leq \frac{717409}{10^6}.$$

Proof. Parameterize the coupled shape by $\mathbb{E}U^2 = \kappa(7/45 + s/15)$, where $\kappa = \mathbb{E}Q^4 - 3 \in [0, 90]$ and $s \in [0, 1]$. For any complete-halfline scalar row with frozen rational parameters $(q_0, q_2, q_4, q_6, \eta)$, the Stein identity and sharp633 theorem imply

$$\mathbb{E}|Q| \leq c + q_0 + q_2 + (3+\kappa)q_4 + [15 + 15\kappa + (K - 633s/15)\kappa^{3/2}]q_6 + \eta\kappa(7/90 + s/30),$$

where $c = \sqrt{2/\pi}$ and $K = 194/\sqrt{10}$. Eight accepted scalar rows are supplemented by two fresh rational Chebyshev rows. Their entire-halfline inequalities are verified by directed compact Taylor and exact polynomial-tail comparisons; a separate direct-Chebyshev implementation replays the new rows with 768-bit sixth-order Taylor bounds on $[0, 80]$ and independently reconstructed Fraction tails. Each row has $q_6 > 0$ and $\eta > 0$, so its price is convex in κ and affine in s on every allowed rectangle. Its validity is global in cumulant, independent of the interval where it was discovered. The exact certificate partitions $[0, 90]$ into 2780 rational intervals, selects one of the ten rows on each, and verifies all four corner prices. The largest outward rational cap is $423485406457/500000000000 = .846970812914 < .847$. An independent MPFR implementation reconstructs all 11120 selected prices and 111200 ten-row candidate checks, verifies source hashes and exact exhaustive coverage, and agrees with the bound. This proves the first assertion using only the freshly included sharp633 and Stein inputs. Normalizing a nonzero cubic projection proves the second by duality. □

Corollary 7. *Let f, g be signs, $h = (f+g)/2$, $k = (f-g)/2$, and $u = \|P_1 h\|_2$. For $T_{.72} = P_1 - (18/25)P_3$, if $u \leq 21/100$, then*

$$\langle f, T_{.72} g \rangle \leq .56307672 < 40/71. \tag{34}$$

Proof. Self-adjointness and orthogonality give

$$\langle f, T_{.72} g\rangle = \|P_1 h\|_2^2 - .72\|P_3 h\|_2^2 - \|P_1 k\|_2^2 + .72\|P_3 k\|_2^2 \le u^2 + .72\rho^2.$$

Since $|k| \le 1$, Theorem 5 applies. At $u = .21$ the exact right side is $441/10000 + 6487209/12500000 = .56307672$. □

The amplitude is the actual first-projection norm, without division by $\sqrt{2/\pi}$. The corollary covers a complete interval of amplitudes, but does not cover the middle range needed for (18).

With the refitted cap $847/1000$, the same elementary energy proof gives $\langle f, T_{.72} g\rangle \le u^2 + (18/25)(847/1000)^2$. It therefore also closes $0 \le u \le 541/2500 = .2164$, where its rational cap is $7042043/12500000 = .56336344 < 40/71$. The frozen regional certificates below retain their original, valid choice $\rho = .849$.

7 A support–source comparison

Let Z be standard Gaussian, independent of a transverse Gaussian vector W. Suppose $|h| + |k| \le 1$, set

$$u = \mathbb{E}[Zh] \ge 0, \qquad p = \mathbb{E}|k|, \qquad r = \|P_3^W \mathbb{E}_Z k\|_2.$$

The source here is the transverse cubic projection. Its independence of Z is essential to the following comparison. For $a > 0$ and $y \ge 0$ write

$$J_a(y) = \mathbb{E}_Z \max\{a|Z|, y\} = ace^{-y^2/(2a^2)} + y\,\mathrm{erf}(y/(\sqrt{2}a)).$$

Proposition 8. *For every $a > 0$ and $\lambda \ge 0$,*

$$au + r + \lambda p \le (1-\rho^2) J_a(\lambda) + \rho^2 J_a(\lambda + 1/\rho), \qquad \rho = 849/1000. \tag{35}$$

Proof. Set $F(y) = J_a(y + \lambda)$ and $x = 1/\rho$. Its third derivative on $[0, \infty)$ is nonpositive:

$$F'''(y) = -\frac{c(y+\lambda)}{a^3} e^{-(y+\lambda)^2/(2a^2)} \le 0.$$

Let $D = F(x) - F(0)$ and interpolate with the quadratic

$$H(y) = F(0) + \beta y + \gamma y^2, \quad \beta = 2D/x - F'(x), \quad \gamma = [xF'(x) - D]/x^2.$$

The Hermite interpolation remainder is $F'''(\xi) y(y-x)^2/6$, so $F(y) \le H(y)$ for every $y \ge 0$. Since F' is increasing and concave, the trapezoid comparison gives $D \ge x(F'(0) + F'(x))/2$, hence $\beta \ge 0$; also $D \le xF'(x)$ implies $\gamma \ge 0$. For any $Y \ge 0$ with $\mathbb{E}Y^2 = 1$ and $\mathbb{E}Y \le \rho$, therefore,

$$\mathbb{E}F(Y) \le F(0) + \beta\rho + \gamma = (1-\rho^2)F(0) + \rho^2 F(1/\rho).$$

For $r > 0$ normalize $P_3^W \mathbb{E}_Z k$ to a variance-one cubic $Q(W)$. Then $\mathbb{E}[Qk] = r$, and pointwise

$$aZh + Qk + \lambda|k| \le \max\{a|Z|, |Q| + \lambda\}.$$

Independence and Theorem 5 allow the preceding comparison with $Y = |Q|$, proving (35). If $r = 0$, choose any normalized transverse cubic: its pairing with k is zero by orthogonality. An unused Gaussian coordinate can be adjoined if necessary. □

At $\lambda = 0$, minimizing the right side of (35) minus au gives the explicit whole-amplitude source cap

$$r \le R_\rho(u) = \begin{cases} \rho, & 0 \le u \le c(1-\rho^2), \\ \rho \operatorname{erf}\sqrt{-\log v}, & c(1-\rho^2) < u < c, \\ 0, & u = c, \end{cases} \qquad v = \frac{u/c - (1-\rho^2)}{\rho^2}. \tag{36}$$

Indeed the derivative of $(1-\rho^2)ac + \rho^2 J_a(1/\rho)$ is $c[1-\rho^2+\rho^2 e^{-1/(2a^2\rho^2)}]$. In the middle range the minimizing value is $a = 1/(\rho\sqrt{-2\log v})$. The two endpoints follow by limits. Retaining $\lambda > 0$ preserves the actual disagreement mass in the comparison. Additional source information is still needed to close the full operator norm bound.

Lemma 9 (Physical clipping of source cells)**.** *Let $c = \sqrt{2/\pi}$, $0 < \rho < 1$, and use the support bound $r \le R_\rho(u)$ of (36). Define*

$$U_\rho(r) = c\{1 - \rho^2 + \rho^2 e^{-[\operatorname{erf}^{-1}(r/\rho)]^2}\}, \qquad 0 < r < \rho,$$

with $U_\rho(0) = c$ and $U_\rho(\rho) = c(1-\rho^2)$. Every physical point in a cell $[u_-, u_+] \times [r_-, r_+]$ also satisfies

$$u \le \min\{u_+, U_\rho(r_-)\}.$$

Thus any outward rational upper bound on $U_\rho(r_-)$ may replace the cell's amplitude endpoint in all native and R3 price maxima. If this upper bound is below u_-, the cell is physically empty.

Proof. The source envelope is decreasing. For $0 < r < \rho$, solving $R_\rho(u) = r$ gives the displayed inverse; its endpoint values follow by limits. Hence $r \ge r_-$ and $r \le R_\rho(u)$ imply the asserted restriction. To certify the inverse without assuming a special-function inverse, bracket $s = \operatorname{erf}^{-1}(r_-/\rho)$ by rational numbers $s_- \le s \le s_+$ using directed monotone evaluations of erf. Since e^{-s^2} decreases for $s \ge 0$, an outward evaluation using s_- gives the required upper bound. □

7.1 Retaining the longitudinal cubic moment

The preceding comparison can also retain $v = \mathbb{E}[H_3(Z)h]$, with $H_3(z) = (z^3 - 3z)/\sqrt{6}$. Take $A > 0$, $B \ge 0$, and write

$$L(z) = Az + Bz^3 = az + dH_3(z), \qquad a = A + 3B, \quad d = \sqrt{6}B, \qquad J_{A,B}(x) = \mathbb{E}_Z \max\{|L(Z)|, |x|\}.$$

Proposition 10. *For every real x,*

$$J_{A,B}(x) \le c\left[A + 2B + \frac{x^2}{2A} - \frac{A+6B}{24A^4}x^4 + \frac{A^2 + 20AB + 120B^2}{240A^7}x^6\right]. \tag{37}$$

If $r > 0$ and $Q = P_3^W \mathbb{E}_Z k/r$ has cumulant κ and shape $V = \mathbb{E}[(\|DQ\|^2/3 - 1)^2]$, then

$$\begin{aligned} au + dv + r \le c\Big[&q_0 + q_2 + (3+\kappa)q_4 \\ &+ q_6\left\{15 + 15\kappa + K\kappa^{3/2} - 633\sqrt{\kappa}(V - 7\kappa/45)\right\}\Big], \end{aligned} \tag{38}$$

where (q_0, q_2, q_4, q_6) are the four coefficients in the brackets of (37).

Proof. For $x \geq 0$ let $z \geq 0$ solve $x = Az + Bz^3$. Differentiating the distribution function gives

$$J'_{A,B}(x) = \Pr(|Z| \leq z), \qquad J''_{A,B}(x) = \frac{ce^{-z^2/2}}{A + 3Bz^2}.$$

Moreover $J_{A,B}(0) = c(A + 2B)$ and $J'_{A,B}(0) = 0$. Set $w = z^2$, $q = x^2 = w(A + Bw)^2$, and $F(q) = J''_{A,B}(\sqrt{q})$. With $D_1 = A + Bw$ and $D_3 = A + 3Bw$,

$$F(q) = ce^{-w/2}/D_3, \qquad \frac{d}{dq} = \frac{1}{D_1 D_3}\frac{d}{dw}.$$

The recurrence

$$P_0 = 1, \qquad P_{j+1} = (P_j/2 - \partial_w P_j)D_1 D_3 + 2jBP_j D_3 + 3(2j + 1)BP_j D_1$$

gives

$$(-1)^j F^{(j)}(q) = \frac{ce^{-w/2}P_j(A, B, w)}{D_1^{2j} D_3^{2j+1}}.$$

For $j = 1, 2, 3$, the expanded numerator has respectively 5, 12 and 22 nonnegative coefficients. The explicit third numerator is given in Appendix C; hence $F'''(q) \leq 0$ on the whole half-line. Taylor's formula with its integral remainder gives

$$F(q) \leq \frac{c}{A} - \frac{c(A + 6B)}{2A^4}q + \frac{c(A^2 + 20AB + 120B^2)}{8A^7}q^2.$$

Integrating twice in x, using the two initial values, proves (37). Evenness covers negative x.

Pointwise $L(Z)h + Qk \leq \max\{|L(Z)|, |Q|\}$, so independence and projection orthogonality give $au + dv + r \leq \mathbb{E}J_{A,B}(Q)$. Apply (37) and Theorem 4. The fourth moment is exact, so its negative coefficient causes no inequality reversal; the sixth coefficient is positive. □

This comparison retains the actual longitudinal cubic moment. A global norm proof must quantify that moment jointly with the same source and profile; assigning it a favorable numerical value would not prove an operator bound.

Proposition 11 (Cubic-head contact source gate)**.** *Let $A > 0$, $B \geq 0$, $L(z) = Az + Bz^3 = az + dH_3(z)$, where $H_3(z) = (z^3 - 3z)/\sqrt{6}$, $a = A + 3B$, $d = \sqrt{6}B$, and let $J(x) = \mathbb{E}_Z \max\{|L(Z)|, |x|\}$. Set $G(s) = J(\sqrt{s})$ on $s \geq 0$. Then $G'' \leq 0$, $G''' \geq 0$ and $G'''' \leq 0$. For every $s_0 > 0$, the cubic Hermite interpolant P matching $G(0), G'(0), G(s_0), G'(s_0)$ satisfies*

$$J(x) \leq P(x^2) = p_0 + p_2 x^2 + p_4 x^4 + p_6 x^6 \quad (x \in \mathbb{R}), \qquad p_6 \geq 0.$$

Here $p_0 = \nu(A+2B)$, $p_2 = \nu/(2A)$, $\nu = \sqrt{2/\pi}$, and, with $D = G(s_0) - p_0 - p_2 s_0$ and $E = G'(s_0) - p_2$,

$$p_4 = (3D - s_0 E)/s_0^2, \qquad p_6 = (s_0 E - 2D)/s_0^3.$$

Any componentwise upward rational rounding $q_j \geq p_j$ gives a global majorant with $q_6 \geq 0$. Its coefficients already contain ν.

Proof. Write $s = w(A + Bw)^2$, $D_1 = A + Bw$, $D_3 = A + 3Bw$. Since $ds/dw = D_1 D_3 > 0$, Gaussian tail differentiation gives

$$G'(s) = \frac{\nu}{2}\int_0^1 \frac{e^{-t^2 w(s)/2}}{A + Bw(s)}\, dt.$$

For $K_\tau(s) = e^{-\tau w(s)}/(A + Bw(s))$, $\tau \geq 0$, differentiation shows

$$K_\tau^{(j)}(s) = (-1)^j \frac{e^{-\tau w} P_j(A, B, w, \tau)}{D_1^{2j+1} D_3^{2j}}, \qquad j = 1, 2, 3.$$

Starting from $P_0 = 1$, the numerator recurrence is

$$P_{j+1} = (\tau P_j - \partial_w P_j) D_1 D_3 + (2j + 1) B P_j D_3 + 6j B P_j D_1.$$

For $j = 1, 2, 3$, exact expansion gives respectively 5, 12 and 22 nonnegative coefficients. Integrating the derivative signs at $\tau = t^2/2$ proves the claimed signs of G. The Hermite interpolation error is

$$G(s) - P(s) = G''''(\xi) s^2 (s - s_0)^2 / 4! \leq 0 \quad (s \geq 0),$$

including outside the interpolation interval. The leading coefficient $p_6 = G[0, 0, s_0, s_0] \geq 0$ by the sign of G'''. □

Corollary 12 (Moment price and retained longitudinal completion)**.** *Let $h = (f + g)/2$, $k = (f - g)/2$, $P_1 h = uZ$, and $v = \mathbb{E}[H_3(Z)h]$. Put $r = \|P_3^W \mathbb{E}_Z k\|_2$ and $Q = P_3^W \mathbb{E}_Z k/r$ when $r > 0$. Write $\kappa = \mathbb{E}Q^4 - 3$ and $\mathbb{E}(\|DQ\|^2/3 - 1)^2 = \kappa(7/45 + \zeta/15)$, with $0 \leq \zeta \leq 1$. The endpoint $\kappa = 0$ is interpreted by continuity. For a majorant from Proposition 11, the physical diamond and the coefficient-633 moment theorem give*

$$au + dv + r \leq A_q(\kappa, \zeta) = q_0 + q_2 + (3 + \kappa) q_4 + \left[15 + 15\kappa + \left(\frac{194}{\sqrt{10}} - \frac{633\zeta}{15}\right) \kappa^{3/2}\right] q_6.$$

No extra ν multiplies this price. For a fixed row the price is convex in κ and affine in ζ. If a certified frame completed its longitudinal penalty by $-bv^2 \leq 2\beta v + \beta^2/b$, with β multiplying normalized H_3, then for any fixed $\lambda \geq 0$ its saved completed bound may be augmented by

$$\lambda \left(A_q - au - r + \frac{\beta d}{b}\right) + \frac{\lambda^2 d^2}{4b}.$$

With λ fixed throughout each price rectangle, the existing cumulant, shape, amplitude and source-chord certification can be reused. If a frame instead stores β_{raw} multiplying $z^3 - 3z$, use $\beta = \sqrt{6}\,\beta_{\mathrm{raw}}$. The same correction is then

$$\lambda \left(A_q - (A + 3B)u - r + \frac{6\beta_{\mathrm{raw}} B}{b}\right) + \frac{3\lambda^2 B^2}{2b}.$$

Both the contact row and λ are chosen once for each whole cell; they cannot be optimized independently at its corner points and then treated as a single convex price.

Proof. Pointwise $L(Z)h + Qk \leq \max(|L(Z)|, |Q|)$, and $\mathbb{E}Qk = r$. Take expectations, use the global majorant, and price its second, fourth and sixth moments. The sixth coefficient is nonnegative and $194/\sqrt{10} - 633/15 > 0$, proving the price properties. Before completing the longitudinal penalty, add the nonnegative quantity $\lambda(A_q - au - dv - r)$ and complete with $(\beta + \lambda d/2)^2/b$ in place of β^2/b. Subtracting the old completion gives the displayed correction. □

For the exact head $A = 26412631/10^7$, $B = 4084047/10^7$ and contact $w_0 = 189/50$, so $s_0 = w_0(A + Bw_0)^2$, one certified rational row is

$$\begin{aligned} q_0 &= 137957132894348330419/(5 \cdot 10^{19}), \\ q_2 &= 1510422344526876849/10^{19}, \\ q_4 &= -163849393411225061/10^{20}, \\ q_6 &= 23608461930109/(25 \cdot 10^{17}). \end{aligned}$$

Its uniform four-corner price on $\kappa \in [2,4]$, $\zeta \in [0,1]$ is at most $29040583649083683/10^{16}$. The exact coefficient-sign proof, rational rounding and directed corner prices are reproduced by `certify_direct_contact_majorant.py`. This source proposition alone does not assert a global operator norm bound. The scalar comparison was independently accepted by an exact rational recurrence check and a separate 768-bit MPFR replay of all 28 upward-rounded coefficients and 98 directed moment prices. A global norm conclusion still requires complete fixed-row, fixed-multiplier cell coverage.

Corollary 13 (A physical source exclusion at the remaining native bottleneck)**.** *In the notation of Proposition 11, every physical source with $u \geq 151/200$, $\kappa \in [2,4]$ and $\zeta \in [0,1]$ satisfies*

$$r \leq \frac{3218564576843929081}{12500000000000000000} < \frac{257485167}{10^9} < \frac{129}{500}.$$

Thus the entire source domain $u \geq .755$, $r \geq .258$, $\kappa \in [2,4]$ is physically impossible. This exclusion is independent of the operator coefficient b and uses no longitudinal-completion multiplier.

Proof. Use the globally valid contact family at $A = 147/50$, $B = 0$, $w_0 = 11/2$, so that $s_0 = 237699/5000$. One upward-rounded rational row is

$$\begin{aligned} q_0 &= 234578060876042414629/10^{20}, \\ q_2 &= 13569465319776621699/10^{20}, \\ q_4 &= -57073674634011611/(5 \cdot 10^{19}), \\ q_6 &= 648580884546013/10^{20} > 0. \end{aligned}$$

These coefficients already contain ν. The price for this single fixed row is convex in κ and affine in ζ. Directed evaluation of the four corners of $[2,4] \times [0,1]$ gives

$$A_q(\kappa,\zeta) \leq \frac{30964814576843929081}{12500000000000000000}.$$

The physical source gate is simply $(147/50)u + r \leq A_q$ since $d = 0$. Subtract $(147/50)(151/200)$ to obtain the asserted bound. All four rational coefficient roundings and four corner prices passed a separate 768-bit MPFR replay; the globally valid contact-family argument also applies at $B = 0$. This is a source-domain exclusion, not a complete operator norm theorem. □

8 Conditional Gaussian profiles and a mixed cubic penalty

Proposition 14 (Sharp conditional Gaussian profile and a quartic budget)**.** *Let Z be standard normal and let measurable real functions h, k satisfy $|h| + |k| \leq 1$. Put*

$$\nu = \sqrt{2/\pi}, \qquad A = \mathbb{E}[Zh], \qquad p = |\mathbb{E}k|, \qquad C = 1 - \nu^2.$$

If $p = 2\Phi(t) - 1$, $t \in [0,\infty]$, then

$$|A| \leq \nu e^{-t^2/2}. \tag{39}$$

This bound is sharp for every p. In particular,

$$A^2 + p^2 - Cp^4 \leq \nu^2. \tag{40}$$

The quartic budget improves the ellipse $A^2 + \nu^2 p^2 \leq \nu^2$ by $Cp^2(1-p^2)$.

Proof. Write $\alpha = |k|$ and $\alpha_0 = \mathbf{1}_{\{|Z|\le t\}}$. Since $\mathbb{E}\alpha \ge p = \mathbb{E}\alpha_0$ and $(|Z| - t)(\alpha - \alpha_0) \ge 0$ pointwise,

$$\mathbb{E}[|Z|\alpha] \ge \mathbb{E}[|Z|\alpha_0].$$

Consequently $|A| \le \mathbb{E}[|Z|(1-\alpha)] \le 2\varphi(t) = \nu e^{-t^2/2}$. Equality is attained by $h = \operatorname{sgn}(Z)\mathbf{1}_{\{|Z|>t\}}$ and $k = \mathbf{1}_{\{|Z|\le t\}}$.

For the polynomial budget set $p(t) = 2\Phi(t) - 1$, so $p' = \nu e^{-t^2/2}$ and $p'' = -tp'$. Define

$$D(t) = \nu^2 e^{-t^2} - \nu^2 + p(t)^2 - Cp(t)^4, \qquad g(t) = p(t) - 2Cp(t)^3 - tp'(t).$$

Then $D' = 2p'g$ and $g' = p'(t^2 - 6Cp^2)$. The ratio $t/p(t)$ strictly increases from $1/\nu$ to infinity: indeed $p(t) > tp'(t)$ for $t > 0$ because p' strictly decreases. Furthermore $1/\nu^2 < 6C$ and $1 - 2C > 0$; for example these follow already from $3 < \pi < 22/7$. Thus g' changes sign exactly once, from negative to positive. As $g(0) = 0$ and $\lim_{t\to\infty} g(t) = 1 - 2C > 0$, the function g has exactly one further zero. It follows that D first decreases and then increases. Its two endpoint limits are zero, so $D \le 0$. Combining this with (39) proves (40). □

Proposition 15 (A mixed cubic completion with one scalar source)**.** *Let Z and W be independent standard Gaussian coordinates. Let f, g be odd signs and put $h = (f+g)/2$, $k = (f-g)/2$,*

$$A(W) = \mathbb{E}_Z[Zh], \qquad k_0(W) = \mathbb{E}_Z k.$$

Suppose $r = \|P_3^W k_0\|_2 > 0$ and write $Q = P_3^W k_0/r = I_3(F)$, with $6\|F\|^2 = 1$. Here Wiener integrals are normalized by $\mathbb{E}I_j(H)^2 = j!\|H\|^2$. Set

$$G = F \otimes_2 F, \quad s = \|G\|^2, \quad D = 1 + \frac{3\|\operatorname{Sym}(F \otimes_1 F)\|^2}{2s}, \quad R_2 = \frac{I_2(G)}{\sqrt{2s}}, \quad \kappa = \mathbb{E}Q^4 - 3.$$

Then $\mathbb{E}R_2 = 0$, $\mathbb{E}R_2^2 = 1$, and

$$\theta := \mathbb{E}[Q^2 R_2] = 18\sqrt{2s}, \qquad \theta^2 = \frac{\kappa}{2D}. \tag{41}$$

For $t = \mathbb{E}[AR_2]$, any $b, w > 0$ and $a, c \in \mathbb{R}$ satisfy

$$\begin{aligned} -bt^2 \le{}& w\{\nu^2 + a^2 + 2ac + c^2(3+\kappa)\} - w\mathbb{E}k_0^2 + wC\mathbb{E}k_0^4 \\ &- 2w\mathbb{E}[(a + cQ^2)A] - \frac{bwc^2\kappa}{2D(b+w)}, \end{aligned} \tag{42}$$

where $\nu^2 = 2/\pi$ and $C = 1 - \nu^2$. The direction $ZR_2(W)$ is a unit cubic Hermite function orthogonal to $H_3(Z) = (Z^3 - 3Z)/\sqrt{6}$. Hence its penalty $-bt^2$ can be retained in addition to a completion using only the longitudinal cubic direction.

Proof. The Gaussian product formula yields

$$P_2(Q^2) = 18I_2(G), \qquad \kappa = 1296s + 1944\|\operatorname{Sym}(F \otimes_1 F)\|^2 = 1296sD.$$

These identities give (41). Also G is positive semidefinite, since $G_{ij} = \sum_{a,b} F_{iab}F_{jab}$. Thus R_2 is the normalized, centered quadratic form determined by this actual cubic source.

Put $\eta = a + cQ^2$. Orthogonal projection onto the unit vector R_2 gives $\mathbb{E}(A - \eta)^2 \ge (t - c\theta)^2$. Completing one scalar square gives

$$bt^2 + w(t - c\theta)^2 \ge \frac{bw}{b+w}c^2\theta^2,$$

and therefore

$$-bt^2 \le w\mathbb{E}(A-\eta)^2 - \frac{bw}{b+w}c^2\theta^2.$$

Conditionally on W, h, k satisfy $|h|+|k|=1$, so Proposition 14 gives $A^2 \le \nu^2 - k_0^2 + Ck_0^4$. Expanding the square and using $\mathbb{E}\eta^2 = a^2 + 2ac + c^2(3+\kappa)$ proves (42). The final orthogonality claim follows from independence of Z, W and $\mathbb{E}R_2 = 0$. □

Insertion into a local scalar criterion. If the original conditional potential contains $2\ell A$ and local moments $\mathbf{k} = (k_0, k_1, k_2, k_3)$, formula (42) replaces

$$\ell \quad \text{by} \quad \ell - w(a + cq^2)$$

at scalar source value q, adds the local polynomial $-wk_0^2 + wCk_0^4$, and adds only the explicit outer price

$$w\{\nu^2 + a^2 + 2ac + c^2(3+\kappa)\} - \frac{bwc^2\kappa}{2D(b+w)}.$$

It introduces no additional source or local moment coordinate. This insertion is an exact inequality, not a global norm certificate: the modified local potential and its source tail must still be bounded on their complete domains. In particular the quadratic dependence on q requires a new chord allowance. If the unmodified local potential is a supremum of affine functions of q, then its modified physical supremum Ψ obeys, for $q \in [q_0, q_1]$,

$$\Psi(q) \le \mathrm{Chord}_{q_0,q_1}\Psi(q) + 2w|c|\nu(q-q_0)(q_1-q).$$

Indeed adding $2w|c|\nu q^2$ makes every candidate convex, because $|A| \le \nu$. Any pre-existing curvature allowance must also be retained.

9 Continuous regional norm certificates

We next convert local inequalities into bounds on complete intervals of the agreement amplitude. All source and cumulant parameters are covered continuously; finite sampling is used only to choose the rational inputs.

Throughout this section the longitudinal Hermites are orthonormal:

$$H_0 = 1, \quad H_1 = Z, \quad H_2 = (Z^2-1)/\sqrt{2}, \quad H_3 = (Z^3 - 3Z)/\sqrt{6}.$$

Section 10 explicitly switches to the raw Hermite convention.

Proposition 16 (A transverse Gram comparison)**.** *Let $b > 0$, let f, g be odd signs, put $h = (f+g)/2$, $k = (f-g)/2$, and rotate so that $P_1 h = uZ$. In the transverse coordinates W, set $k_j(W) = \mathbb{E}_Z[H_j(Z)k]$, $h_j(W) = \mathbb{E}_Z[H_j(Z)h]$, and $r = \|P_3^W k_0\|_2$. Suppose the even block E on coordinates $(0,2)$ and odd block R on coordinates $(1,3)$ satisfy*

$$E \succeq 0, \quad E - \mathrm{diag}(-1, b) \succeq 0, \quad R \succeq 0, \quad R - \mathrm{diag}(-1, b) \succeq 0, \quad R - \mathrm{diag}(b, 0) \succeq 0, \qquad E_{22} > 0.$$

Let M be the corresponding four-coordinate matrix. For arbitrary real $\ell, \beta, \delta, \omega$, define

$$s = (\delta, 0, -\omega, 0), \qquad \xi = \frac{(\omega - E_{02})^2}{E_{22}} - E_{00} + b - 2\delta.$$

Let $\mathcal{P}(t)$ be the supremum, over conditional functions satisfying $|h| + |k| \le 1$, of

$$\mathbf{k}^T M \mathbf{k} + 2t\, s^T \mathbf{k} + 2\ell h_1 + 2\beta h_3 + \beta^2/b, \qquad \mathbf{k} = (k_0, k_1, k_2, k_3)^T.$$

For $r > 0$, with $Q = P_3^W k_0 / r$, one has

$$\langle f, T_b g \rangle \le u^2 - 2\ell u + \mathbb{E}\mathcal{P}(rQ) + \xi r^2. \tag{43}$$

For $r = 0$, replace the expectation by $\mathcal{P}(0)$.

Proof. Retain $-b(\mathbb{E}h_3)^2$ from the agreement penalty and use $-bx^2 \le 2\beta x + \beta^2/b$. Expand the disagreement function by longitudinal and transverse Hermite degree. In transverse degree one, the even block must dominate $\mathrm{diag}(-1, b)$; in transverse degrees zero and two, the odd block must dominate the two displayed diagonal matrices. In the source direction Q, write $a = \mathbb{E}[Qk_2]$. The even quadratic and source terms are

$$E_{22}a^2 + 2(E_{02} - \omega)ra + (E_{00} + 2\delta + \xi)r^2 \ge br^2,$$

by their exact Schur completion. The component of $P_3^W k_0$ orthogonal to Q is zero by definition. All remaining transverse components are dominated by the positive blocks. Finally insert $2\ell\mathbb{E}Zh - 2\ell u = 0$. These comparisons give (43) after conditional maximization. □

For completeness, the finite local comparison is particularly simple. If the actual conditional moment vector lies in a box centered at c with half widths w, exact quadratic completion and the diamond constraint give the entire box bound

$$\mathcal{L}(t; h, k) \le 2\mathbb{E}_Z \max\{|(Mc + ts) \cdot H|, |\ell H_1 + \beta H_3|\} - c^T M c + \beta^2/b + w^T |M| w, \tag{44}$$

where $\mathcal{L}$ denotes the objective defining $\mathcal{P}$ and $|M|$ is entrywise absolute value. This is a bound for actual moments in that box, with no stationary-point assumption. Bessel's inequality and $|k_j| \le \mathbb{E}|H_j|$ restrict the moments to

$$\sum_{j=0}^{3} k_j^2 \le 1, \qquad (|k_0|, |k_1|, |k_2|, |k_3|) \le (1, 4/5, 7/10, 5/8).$$

The expectation in (44) is the sum of two Gaussian absolute-cubic integrals, since $\max(|A|, |B|) = (|A + B| + |A - B|)/2$. Certified real-root isolation and Gaussian antiderivatives evaluate these integrals with outward rounding.

The potential $\mathcal{P}$ is even and convex in t. Its certified nodal heights therefore give upper chords on every source interval; a tail slope at least $2\|s\|$ gives an affine extension to infinity. Denote this even convex upper function by F. To price $\mathbb{E}F(rQ)$, solve $g_F'(x) - x g_F(x) = F(rx) - \mathbb{E}F(rN)$ for a standard normal N. For any $\lambda \ge 0$, any odd regression ψ, and $\eta > 0$, the same Stein calculation as in Section 6 gives

$$\mathbb{E}F(rQ) \le \mathbb{E}F(rN) + \lambda(\rho - c) + \mathbb{E}\left[\psi'(Q) - Q\psi(Q) + J_\eta(g_F'(Q) - \lambda g'(Q) - \psi'(Q))\right] + \frac{\eta}{2}\mathbb{E}U^2, \tag{45}$$

$$J_\eta(x) = \begin{cases} x^2/(2\eta), & x \le \eta, \\ x - \eta/2, & x \ge \eta. \end{cases}$$

Here $c = \sqrt{2/\pi}$, g is the absolute-value Stein solution above, and $\rho = 849/1000$. The clipped quadratic follows by maximizing $-vx - \eta v^2/2$ over $v \ge -1$, since $U \ge -1$. Smooth approximation justifies the finitely many target knots.

A whole-line sextic majorant for the bracket in (45), with nonnegative sextic coefficient, has a price convex in κ and affine in the shared variance shape. Consequently its four endpoint comparisons cover each complete cumulant–shape rectangle. Source interpolation adds only

$$(-\xi)_+ \max_i (r_{i+1} - r_i)^2/4$$

to the endpoint prices, because $\mathbb{E}F(rQ)$ is convex in r. Checking the two endpoints of $u^2 - 2\ell u$ covers an entire amplitude interval. These facts explain the continuous scope of the following certificates.

Theorem 17 (Complete regional amplitude bounds)**.** *For odd Gaussian signs f, g, put $u = \|P_1((f+g)/2)\|_2$. In every finite Gaussian dimension,*

$$\langle f, T_{.72} g\rangle < \begin{cases} 563367/10^6, & 0 \le u \le 341/500, \\ 563334/10^6, & 77/100 \le u \le \sqrt{2/\pi}. \end{cases} \qquad < \frac{40}{71}.$$

Proof. The seven frozen profiles in Appendix D cover the intervals $[.14, .42]$, $[.30, .485]$, $[.40, .54]$, $[.50, .638]$, $[.60, .662]$, $[.662, .682]$, and $[.77, .797885]$. Each satisfies Proposition 16, complete local moment-box comparisons, and complete scalar inequalities of (45). Directed prices and exact convex endpoint comparisons cover every source amplitude, cumulant and shared shape in each region. Their first six rational caps are all at most $563367/10^6$; the seventh cap is $563334/10^6$. For $u \le .14$, the energy comparison gives $u^2 + .72\rho^2 \le .53857672$. The intervals therefore cover every $u \le .682$. Finally $u \le \mathbb{E}|Z| = \sqrt{2/\pi} < .797885$, so the last interval contains every feasible $u \ge .77$. □

The accepted broad profile uses a source grid ending exactly at ρ, which is valid at every agreement amplitude by projection norm duality. Hence its original discovery interval imposes no restriction on its larger accepted interval. The other extensions retain their verified outward source endpoint. This accounts for the continuous scope of the theorem. The remaining entire amplitude range is $.682 < u < .77$; a local source-parameter subregion inside that range does not complete its full norm cover.

10 Retaining the actual cubic projection

Let $T_b = P_1 - bP_3$, $b > 0$, on real standard Gaussian space, where P_j is the orthogonal projection onto total Hermite degree j. For odd signs f, g put $h = (f+g)/2$, $k = (f-g)/2$. Then (h, k) takes the four values $(\pm 1, 0), (0, \pm 1)$. Rotate the Gaussian coordinates as (Z, W) so that $P_1 h = uZ$, with Z independent of W. Use the raw longitudinal Hermites

$$H_0 = 1, \quad H_1 = Z, \quad H_2 = Z^2 - 1, \quad H_3 = Z^3 - 3Z, \qquad m_j(W) = \mathbb{E}_Z[H_j k].$$

If $r = \|P_3^W m_0\|_2 > 0$, define the normalized cubic source $Q = P_3^W m_0/r$. Thus $\mathbb{E}Q^2 = 1$ and $\mathbb{E}[Qm_0] = r$. All remaining projections act on W.

Proposition 18 (An exact residual of the physical cubic source)**.** *Put*

$$\kappa = \mathbb{E}Q^4 - 3, \quad c_6 = \mathbb{E}Q^6 - 15\mathbb{E}Q^4 + 30, \quad S(q) = q^3 - 3q,$$

$$\mathcal{R} = Q^3 - (3+\kappa)Q - P_1^W(Q^3), \quad \Delta = \mathcal{R} - S(Q), \quad D = \kappa^2 + \|P_1^W(Q^3)\|_2^2.$$

The residual $\mathcal{R}$ is odd, orthogonal to first chaos and to Q, and

$$\mathbb{E}\mathcal{R}^2 = 6 + 9\kappa + c_6 - D, \quad \mathbb{E}\Delta^2 = D, \quad \mathbb{E}[Q\Delta] = -\kappa, \quad \mathbb{E}[S(Q)\Delta] = -D. \tag{46}$$

Proof. Since Q has pure degree three, $P_1^W(Q^3)$ is orthogonal to Q. The coefficient removed from Q^3 along the unit vector Q is $\mathbb{E}Q^4 = 3 + \kappa$. Also $\Delta = -\kappa Q - P_1^W(Q^3)$. Orthogonal expansion of these two expressions proves all four identities. □

Write $Q = I_3(F)$ and put $s_F = \|F \otimes_2 F\|_F^2$ and $z = 1296 s_F/\kappa \in [2/5, 1]$, as in the tensor shape comparison of Section 5. The residual parameter has the dimension-free bounds

$$\kappa^2 \le D \le \kappa^2 + \frac{9}{4}(\kappa z)^{3/2}. \tag{47}$$

Indeed Gaussian differentiation and the product formula give $\mathbb{E}[X_i Q^3] = 324\langle F_i, F \otimes_2 F\rangle$. The flattening map of F has squared operator norm at most $\sqrt{s_F}$, so the sum of the squared inner products is at most $s_F^{3/2}$. Multiplication by 324^2 proves (47).

Proposition 19 (An inverse-free physical dual criterion)**.** *Fix $p = (p_0, p_2)$ and a symmetric matrix $R = \left(\begin{smallmatrix} R_{11} & R_{13} \\ R_{13} & R_{33} \end{smallmatrix}\right)$ such that*

$$p_2^2 \ge \frac{b}{2}(1 + p_0^2), \qquad R \succeq 0, \quad R - \operatorname{diag}(-1, b/6) \succeq 0, \quad R - \operatorname{diag}(b, 0) \succeq 0. \tag{48}$$

Choose real $\ell, \beta, \gamma, y, \zeta, c, d$ and $G > 0$. Write

$$\sigma^2 = p_0^2 + 2p_2^2, \quad \rho = 1 + \zeta^2/G, \quad \mathfrak{g}(q) = cq + dS(q), \quad a(q) = \zeta[-S(q) + \mathfrak{g}(q)/(2G)].$$

For $v = (s, v_1, v_3)$ define

$$\begin{aligned} L(z) &= \ell z + \beta H_3(z), \\ B(q, z; v) &= q\gamma + q(y - p_0 p_2 r)H_2(z) + [a(q) + \rho s][p_0 + p_2 H_2(z)] \\ &\quad + (R_{11}v_1 + R_{13}v_3)z + (R_{13}v_1 + R_{33}v_3)H_3(z), \end{aligned}$$

and the physical dual ellipsoid

$$\mathcal{E}_p = \{(s, v_1, v_3) : s^2/\sigma^2 + v_1^2 + v_3^2/6 \le 1\}.$$

Put

$$\Phi(q) = \sup_{v \in \mathcal{E}_p} \left\{ 2\mathbb{E}_Z \max(|L(Z)|, |B(q, Z; v)|) - \rho s^2 - R_{11}v_1^2 - 2R_{13}v_1 v_3 - R_{33}v_3^2 + \frac{\mathfrak{g}(q)^2}{4G} \right\}. \tag{49}$$

Then

$$\begin{aligned} \langle T_b f, g\rangle \le u^2 - 2\ell u + \frac{6\beta^2}{b} + (b - p_0^2)r^2 - 2\gamma r + \frac{y^2}{p_2^2} + \mathbb{E}\Phi(Q) \\ + \zeta^2(6 + 9\kappa + c_6) - c\kappa + (G - d - \zeta^2)D. \end{aligned} \tag{50}$$

The ellipsoid in (49) *may be restricted to $v_1 \ge 0$ without changing the supremum, and $\Phi(-q) = \Phi(q)$.*

Proof. Let $V = p_0 m_0 + p_2 m_2$ and $t = \mathbb{E}[Q m_2]$. Let Π project onto odd transverse degrees at least three, orthogonally to Q. Hermite orthogonality and (48) give

$$-\|P_1 k\|_2^2 + b\|P_3 k\|_2^2 \le \mathbb{E}V^2 + \mathbb{E}(R_{11} m_1^2 + 2R_{13} m_1 m_3 + R_{33} m_3^2) + (b - p_0^2) r^2 - 2p_0 p_2 r t - p_2^2 t^2 - \|\Pi V\|_2^2.$$

Indeed the relevant raw-Hermite blocks are $\mathrm{diag}(-1, b/2)$ on (m_0, m_2) in transverse degree one, $\mathrm{diag}(-1, b/6)$ on (m_1, m_3) in degree zero, and $\mathrm{diag}(b, 0)$ on that latter pair in degree two. The first gate is exactly the first block inequality for pp^T. In even longitudinal/transverse degree three, $P_3^W m_0 = rQ$; subtracting the Q component of V leaves precisely the indicated unused loss. Higher odd transverse degrees supply the remaining part of $\|\Pi V\|_2^2$.

Now $\mathcal{R}$ lies in the range of Π, so

$$-\|\Pi V\|_2^2 \le -2\zeta \mathbb{E}[\mathcal{R} V] + \zeta^2 \mathbb{E}\mathcal{R}^2.$$

The pointwise square inequality

$$-2\zeta \Delta V \le \mathfrak{g}(Q)\Delta + G\Delta^2 + (\zeta V + \mathfrak{g}(Q)/2)^2 / G$$

and (46) yield the last line of (50); the remaining local quadratic is ρV^2 with linear coefficient $2a(Q)V$ and constant $\mathfrak{g}(Q)^2/(4G)$. Complete $-p_2^2 t^2 \le 2yt + y^2/p_2^2$, and insert the exact zeros $2\gamma \mathbb{E}[Q m_0] - 2\gamma r$ and $2\ell \mathbb{E}[Z h] - 2\ell u$. Finally $-b\|P_3 h\|_2^2 \le -(b/6)(\mathbb{E}[H_3 h])^2 \le 2\beta \mathbb{E}[H_3 h] + 6\beta^2/b$.

For each fixed W, Bessel's inequality gives $m_0^2 + m_1^2 + m_2^2/2 + m_3^2/6 \le \mathbb{E}_Z k^2 \le 1$. Hence $(V, m_1, m_3) \in \mathcal{E}_p$ by Cauchy–Schwarz in the even block. For a positive quadratic form A, the identity

$$x^T A x = \sup_v \{2v^T A x - v^T A v\}$$

holds also when A is singular, with equality at $v = x$. Apply it to $\mathrm{diag}(\rho, R)$ and interchange the two suprema. Maximizing the resulting linear functional over the four pointwise possibilities for (h, k) gives $2\max(|L|, |B|)$. Restricting the dual supremum to $\mathcal{E}_p$ loses no value because it contains every physical moment vector. Conversely every dual vector gives at most the physical supremum by the same quadratic identity. This proves (49) without a matrix inverse.

Under $z \mapsto -z$, the simultaneous reflection $(v_1, v_3) \mapsto (-v_1, -v_3)$ preserves the value and the ellipsoid. Thus $v_1 \ge 0$ suffices. Under $(q, s) \mapsto (-q, -s)$ the even part of B changes sign; reflecting z and changing the overall sign of B restores the same absolute value. This proves the evenness in q. □

Proposition 20 (A full even-block criterion)**.** *Retain the physical residual-source notation, including $m_j = \mathbb{E}_Z[H_j k]$. Let $E = (E_{ij})_{i,j\in\{0,2\}}$, $R = (R_{ij})_{i,j\in\{1,3\}}$ be symmetric matrices and $p = (p_0, p_2)^T$ satisfying*

$$E \succeq 0, \qquad E - \mathrm{diag}(-1, b/2) \succeq 0, \qquad E - pp^T \succeq 0,$$
$$R \succeq 0, \qquad R - \mathrm{diag}(-1, b/6) \succeq 0, \qquad R - \mathrm{diag}(b, 0) \succeq 0.$$

In particular $E_{22} \ge b/2 > 0$. For arbitrary real $\ell, \beta, \gamma, y, \zeta, c, d$ and $G > 0$, put

$$S(q) = q^3 - 3q, \quad \mathfrak{g}(q) = cq + dS(q),$$
$$a(q) = \zeta[-S(q) + \mathfrak{g}(q)/(2G)], \quad L(z) = \ell z + \beta H_3(z).$$

With the coordinate order $(0,1,2,3)$, let N have even block $E+(\zeta^2/G)pp^T$ and odd block R, and set

$$b_0(q)=(q\gamma+a(q)p_0,\,0,\,q(y-E_{02}r)+a(q)p_2,\,0)^T,\quad H(z)=(H_0,H_1,H_2,H_3)^T.$$

Define the physical ellipsoid and local dual value by

$$\mathcal{E}=\{v:v_0^2+v_1^2+v_2^2/2+v_3^2/6\le 1\},$$
$$\Phi_E(q)=\sup_{v\in\mathcal{E}}\left\{2\mathbb{E}_Z\max\big(|L(Z)|,|(b_0(q)+Nv)^TH(Z)|\big)-v^TNv+\frac{\mathfrak{g}(q)^2}{4G}\right\}.$$

Then

$$\langle T_bf,g\rangle\le u^2-2\ell u+\frac{6\beta^2}{b}+(b-E_{00})r^2-2\gamma r+\frac{y^2}{E_{22}}+\mathbb{E}\Phi_E(Q)$$
$$+\zeta^2(6+9\kappa+c_6)-c\kappa+(G-d-\zeta^2)D.$$

The supremum may be restricted to $v_1\ge 0$, and Φ_E is even.

Proof. Put $t=\mathbb{E}[Qm_2]$, $V=p_0m_0+p_2m_2$, and let Π project onto odd transverse degrees at least three, orthogonally to Q. Hermite orthogonality gives

$$-\|P_1k\|_2^2+b\|P_3k\|_2^2\le\mathbb{E}(m_0,m_2)E(m_0,m_2)^T+\mathbb{E}(m_1,m_3)R(m_1,m_3)^T$$
$$+(b-E_{00})r^2-2E_{02}rt-E_{22}t^2-\|\Pi V\|_2^2.$$

To verify this inequality, the transverse degree-one even block is $\mathrm{diag}(-1,b/2)$, while the degree-zero and degree-two odd blocks are $\mathrm{diag}(-1,b/6)$ and $\mathrm{diag}(b,0)$. The corresponding gates cover them. In transverse degree three, $P_3^Wm_0=rQ$ exactly. Subtracting the Q-component of the E quadratic leaves a nonnegative multiple $E_{22}\|P^W_{3,Q^\perp}m_2\|_2^2$. This covers the same component of $\|\Pi V\|_2^2$, since $E_{22}\ge p_2^2$. In every higher odd transverse degree, $E-pp^T\succeq 0$ covers the unused loss. The other odd-block degrees use $R\succeq 0$.

Apply the residual square completion to V exactly as before: $-\|\Pi V\|_2^2\le -2\zeta\mathbb{E}[\mathcal{R}V]+\zeta^2\mathbb{E}\mathcal{R}^2$, followed by the completion of $-2\zeta\Delta V$. This adds only $(\zeta^2/G)pp^T$ to the even block and gives the displayed D,c_6,κ terms. Complete $-E_{22}t^2\le 2yt+y^2/E_{22}$ and the longitudinal agreement penalty, and insert the exact linear and source identities. For each transverse point, Bessel's inequality places the physical four-moment vector in $\mathcal{E}$. The identity $x^TNx=\sup_v(2v^TNx-v^TNv)$, valid also for singular positive semidefinite N, and the four pointwise values of (h,k) now give precisely Φ_E. The reflections $z\mapsto -z$ and $(q,v_0,v_2)\mapsto(-q,-v_0,-v_2)$ give the final symmetry claims. □

Transport and a necessary concavity allowance. Set $y(r)=y_0+E_{02}(r-r_0)$ and $v_y=y_0-E_{02}r_0$. This freezes b_0 and hence the whole local certificate, while

$$(b-E_{00})r^2-2\gamma r+\frac{y(r)^2}{E_{22}}=a_rr^2-2\left(\gamma-\frac{E_{02}v_y}{E_{22}}\right)r+\frac{v_y^2}{E_{22}},\qquad a_r=b-E_{00}+\frac{E_{02}^2}{E_{22}}.$$

Unlike the rank-one case, a_r can be negative. On a source cell $r\in[r_-,r_+]$, endpoint interpolation remains an upper bound after adding $\max(0,-a_r)(r_+-r_-)^2/4$. No compact local certificate, or fixed-source price, alone proves a global operator norm bound.

Proposition 21 (Jensen reductions with a rank local oracle)**.** *Retain the raw-Hermite rank residual-source notation $p = (p_0, p_2)$, L, B, $\mathcal{E}_p$, Φ, $b > 0$ and R. Let $0 \le \omega < 1$, $\nu^2 = 2/\pi$, and put $C_0 = 1 - \nu^2$. Use the exact source transport*

$$y(r) = y_0 + p_0 p_2 (r - r_0), \qquad v_y = y_0 - p_0 p_2 r_0, \qquad \eta = \gamma - p_0 v_y / p_2.$$

Write $\mathcal{C}$ for the usual residual price, including $6\beta^2/b$ and v_y^2/p_2^2, but excluding the amplitude and source quadratics and $\mathbb{E}\Phi(Q)$. Assume the three usual odd-block spectral gates for R.

For the ellipse reduction choose any $v_{\mathrm{e}} \ge \nu^2$ and assume only

$$p_2^2 \ge \frac{b}{2}\left(1 + \frac{p_0^2}{1 + v_{\mathrm{e}}\omega}\right). \tag{51}$$

Then the ordinary rank local oracle, with constant longitudinal head, gives

$$\langle T_b f, g\rangle \le (1-\omega)u^2 - 2\ell u + (b - v_{\mathrm{e}}\omega)r^2 - 2\eta r + \mathcal{C} + v_{\mathrm{e}}\omega + \mathbb{E}\Phi(Q). \tag{52}$$

The stronger old gate $p_2^2 \ge (b/2)(1 + p_0^2)$ is not required.

For the quartic reduction replace the even gate by

$$p_2^2 \ge \frac{b}{2}\left(1 + \frac{p_0^2}{1 + \omega}\right). \tag{53}$$

Define the modified local oracle by adding one independent bounded dual variable $\tau \in [-1, 1]$ to the rank oracle: replace $B(q, Z; v)$ inside $2\mathbb{E}_Z \max(|L|, |B|)$ by $B(q, Z; v) + 2\omega C_0 \tau^3$, subtract $3\omega C_0 \tau^4$ outside this expectation, and maximize over both $v \in \mathcal{E}_p$ and $\tau \in [-1, 1]$. Denote the result by $\Phi_\omega(q)$. Then

$$\langle T_b f, g\rangle \le (1-\omega)u^2 - 2\ell u + (b - \omega)r^2 - 2\eta r + \mathcal{C} + \omega\nu^2 + \mathbb{E}\Phi_\omega(Q), \tag{54}$$

and $0 \le \Phi_\omega(q) - \Phi(q) \le \omega C_0$. Neither extension introduces a quadratic dependence of the longitudinal head on q. Existing residual curvature allowances remain necessary; the quartic extension adds no new such allowance and its coarse tail increases by at most ωC_0.

Proof. Set $A(W) = \mathbb{E}_Z[Zh]$ and $m_0(W) = \mathbb{E}_Z k$. Alignment of $P_1 h = uZ$ gives $\mathbb{E}A = u$, and Jensen gives $\mathbb{E}A^2 \ge u^2$. The conditional profile budget proves

$$A^2 + \nu^2 m_0^2 \le \nu^2, \qquad A^2 + m_0^2 - C_0 m_0^4 \le \nu^2.$$

Thus the integrated quantities

$$v_{\mathrm{e}} - u^2 - v_{\mathrm{e}}\mathbb{E}m_0^2, \qquad \nu^2 - u^2 - \mathbb{E}m_0^2 + C_0\mathbb{E}m_0^4$$

are nonnegative, using $v_{\mathrm{e}} \ge \nu^2$ for the first.

Apply the full even-block criterion with $E = pp^T + a e_0 e_0^T$, taking $a = v_{\mathrm{e}}\omega$ or $a = \omega$. The gates $E \succeq 0$ and $E - pp^T \succeq 0$ are automatic. The remaining gate is equivalent to

$$\det(E - \mathrm{diag}(-1, b/2)) = (1 + a)p_2^2 - \frac{b}{2}(1 + a + p_0^2) \ge 0,$$

which is exactly the respective displayed even gate. In particular $p_2 \ne 0$. Add ω times the corresponding nonnegative integrated budget to the full criterion. Its negative m_0^2 term cancels the extra local even-block entry exactly, so the effective local quadratic remains the original rank

quadratic. The extra outer terms are $\omega(v_{\mathrm{e}} - u^2 - r^2 v_{\mathrm{e}})$ in the ellipse case and $\omega(\nu^2 - u^2 - r^2)$ in the quartic case. Exact $y(r)$ transport then gives the two stated prices.

In the quartic case the additional local term is $\omega C_0 m_0^4$. Convexity of x^4 gives, for $|x| \le 1$,

$$x^4 = \sup_{|\tau| \le 1} (4\tau^3 x - 3\tau^4),$$

with equality at $\tau = x$. Since $m_0 = \mathbb{E}_Z k$, the linear part shifts the support field by $2\omega C_0 \tau^3$, proving the stated local formula by the same PSD duality as for the rank oracle. The actual physical moment vector and $\tau = m_0$ are admissible, and the two dual suprema may be interchanged with the physical supremum. The added physical term lies between zero and ωC_0, proving the local gap and tail statements. The extra dual terms do not depend on q, and $1 - \omega > 0$ preserves the positive amplitude curvature needed for exact polygon pricing. □

These are analytic sufficient criteria. Any newly fitted parameters, including ones violating the stronger old rank gate, require validation of the true even-block gates and a complete local scalar certificate before yielding an operator norm bound.

Proposition 22 (A frozen-centering scalar majorant)**.** *Let $U = \|\nabla_W Q\|^2/3 - 1$ and $\mathcal{V} = \mathbb{E}U^2$. Fix any $\theta \in \mathbb{R}$, $\eta > 0$, real a_0, a_2, and a continuously differentiable test function φ such that φ and φ' have at most polynomial growth. Set*

$$e(q) = \varphi'(q) - a_0 - a_2(q^2 - 1), \qquad \lambda(q) = -1 - \theta(q^2 - 1),$$

$$P(q) = \begin{cases} e(q)^2/(2\eta), & e(q) \ge \eta\lambda(q), \\ \lambda(q)e(q) - \eta\lambda(q)^2/2, & e(q) < \eta\lambda(q). \end{cases}$$

If, for every real q,

$$\Phi(q) \le A_0 + A_2 q^2 + A_4 q^4 + A_6 q^6 + q\varphi(q) - [1 + \theta(q^2 - 1)]\varphi'(q) - P(q), \tag{55}$$

then

$$\begin{aligned} \mathbb{E}\Phi(Q) \le{}& A_0 + A_2 + (3 + \kappa)A_4 + (15 + 15\kappa + c_6)A_6 \\ &+ \frac{\eta}{2}[\mathcal{V} - 2\theta\kappa/3 + \theta^2(\kappa + 2)] + a_2[\kappa/3 - \theta(\kappa + 2)]. \end{aligned} \tag{56}$$

In particular one may freeze $\theta = \kappa_0/[3(\kappa_0 + 2)]$ throughout a whole cumulant interval; the covariance correction in (56) *keeps its displayed positive sign. When φ is odd, the proposed majorant is even, so it suffices to verify* (55) *for $q \ge 0$. The same proof applies to the full-even-block potential Φ_E after replacing Φ by Φ_E.*

Proof. Since Q has pure degree three, Gaussian integration by parts gives $\mathbb{E}[Q\varphi(Q)] = \mathbb{E}[(1 + U)\varphi'(Q)]$. It also gives $\mathbb{E}U = 0$ and $\mathbb{E}[U(Q^2 - 1)] = \kappa/3$; the latter follows by using $\varphi(q) = q^3$. Put $U_\theta = U - \theta(Q^2 - 1)$. Then $U_\theta \ge \lambda(Q)$, its variance is $\mathcal{V} - 2\theta\kappa/3 + \theta^2(\kappa + 2)$, and $\mathbb{E}[U_\theta(Q^2 - 1)] = \kappa/3 - \theta(\kappa + 2)$. By construction $P(q) = \sup_{v \ge \lambda(q)}[ve(q) - \eta v^2/2]$. Therefore $U_\theta e(Q) - P(Q) \le \eta U_\theta^2/2$. Take expectations in (55) and apply these identities. This proves (56). □

Exact transport of the source amplitude. Freeze a reference r_0 and y_0, and set

$$y(r) = y_0 + p_0 p_2 (r - r_0), \qquad v_y = y_0/p_2 - p_0 r_0.$$

Then $y(r) - p_0 p_2 r = y_0 - p_0 p_2 r_0$, so the entire function Φ is unchanged for every r. Its outer quadratic simplifies exactly to

$$(b - p_0^2) r^2 - 2\gamma r + y(r)^2/p_2^2 = b r^2 - 2(\gamma - p_0 v_y) r + v_y^2. \tag{57}$$

Consequently no local amplitude error is needed for this transport; both the u and r outer quadratics are convex and are bounded on rectangles by their vertex values. The source moments and cumulants must still be covered separately. A proposed scalar majorant becomes a norm certificate only after it bounds (49) for the whole real q line and its outer expectation is bounded for every admissible physical source. No finite grid maximization establishes those requirements. The case $r = 0$ can be handled separately by a source-free criterion, without normalizing a zero projection.

Proposition 23 (A certified cubic-source rectangle)**.** *For $b = 18/25$, every odd Gaussian sign pair whose physical parameters satisfy*

$$u \in [7549/10000, 7551/10000], \qquad r \in [2574/10000, 2577/10000], \qquad \kappa \in [51/100, 55/100]$$

obeys

$$\langle (P_1 - bP_3) f, g \rangle \le \frac{28167}{50000} = 0.56334 < \frac{40}{71}.$$

The bound covers every admissible cubic tensor shape and every Gaussian dimension.

Certificate proof. The exact rational profile is frozen in `frozen_R3_profile_exact.json`. The scalar majorant uses reserves $2 \cdot 10^{-5}$ in its constant coefficient and 10^{-6} in its degree-six coefficient. Exact rational Bernstein derivative bounds and directed 256-bit MPFR arithmetic certify the whole interval $q \in [0, 8]$: the completed partition has 6751 nodes and 3376 accepted leaves. A fresh independent traversal rechecked all accepted leaves and the physical-center quadratic radii, with stable source hashes. The independently replayed $[8, 64]$ cover and exact nonnegative-coefficient tail proof for $q \ge 64$ remain valid after increasing the constant reserve. Evenness covers negative q.

Use the fresh sixth-cumulant comparison and the exact cubic projection bound

$$c_6 \le \left[\frac{194}{\sqrt{10}} - \frac{633}{9}(z - 2/5) \right] \kappa^{3/2}, \qquad D \le \kappa^2 + \frac{9}{4}(\kappa z)^{3/2}, \quad z \in [2/5, 1].$$

In the outer price their respective coefficients are nonnegative. After the exact $y(r)$ transport, the price is separately convex in u, r, κ, z; thus its 16 rectangle vertices suffice. Independent directed evaluation gives the upper bound $0.563333433603497\ldots < 28167/50000$. □

This certifies the displayed rectangle of physical parameters. It is not a global bound for $\|P_1 - (18/25)P_3\|_{\infty \to 1}$.

Proposition 24 (Certified moving bands of cubic-source parameters)**.** *Let $b = 18/25$ and use the physical parameters u, r, κ of an odd Gaussian sign pair. Define*

$$t = r - \frac{5151}{20000} + 3\left(u - \frac{151}{200}\right).$$

In each of the following three cases, the pairing is bounded by the listed cap, in every Gaussian dimension and for every admissible dimensionless cubic tensor shape $z \in [2/5, 1]$*:*

u	$\lvert t\rvert$ *at most*	κ	$\langle (P_1 - bP_3)f, g\rangle$ *at most*
$[.7538, .7562]$	.00025	$[.3, .8]$	28168/50000
$[.75, .76]$	.001	$[.3, .8]$	451/800
$[.7505, .7595]$	.001	$[.15, 1.1]$	451/800

All finite decimals in the table are exact rational numbers. The first cap is $0.56336 < 40/71$*; the other two are* 0.56375*.*

Certificate proof. Use the accepted whole-line scalar majorant from the exact frozen profile, with constant reserve $1/50000$ and sixth-degree reserve $1/1000000$. The transport $y(r) = y_0 + p_0 p_2 (r - r_0)$ leaves this entire local majorant unchanged. Its outer amplitude part is

$$u^2 - 2\ell u + br^2 - 2(\gamma - p_0 v_y)r + v_y^2, \qquad v_y = y_0/p_2 - p_0 r_0.$$

After $r = 5151/20000 - 3(u - 151/200) + t$, its Hessian divided by two is

$$\begin{pmatrix} 187/25 & -54/25 \\ -54/25 & 18/25 \end{pmatrix}, \qquad \det = 18/25 > 0.$$

Thus it is jointly convex in (u, t).

In the source-moment price the coefficients of the sixth cumulant and the projection allowance are respectively

$$A = \frac{22262234067601129}{2500000000000000000000} > 0, \qquad B = \frac{172316040932398871}{2500000000000000000000} > 0.$$

The fresh source bounds therefore give a price consisting of affine terms in each source coordinate, plus

$$\left[A \left\{ \frac{194}{\sqrt{10}} - \frac{633}{9}(z - 2/5) \right\} + \frac{9}{4} B z^{3/2} \right] \kappa^{3/2} + B\kappa^2.$$

The coefficient in braces is positive for $2/5 \le z \le 1$. Consequently the price is separately convex in κ and z. Each affine parameter rectangle (u, t, κ, z) is therefore bounded by its 16 vertex prices. Independent exact-profile directed evaluation at 256 bits gives, respectively,

$$0.563356280368, \qquad 0.563648547036, \qquad 0.563686934863$$

as rational upper bounds for the three vertex maxima. These are strictly below the respective table caps. □

The restrictions on t are restrictions on the actual projection norm r. These moving bands do not cover every r at their listed amplitudes and do not by themselves bound the global operator norm.

Proposition 25 (Further certified source regions)**.** *For* $b = 18/25$*, use the physical parameters* u, r, κ *of an odd Gaussian sign pair. In every Gaussian dimension and for every admissible cubic tensor shape,*

$$u \in [341/500, 1381/2000], \quad 0 \le \kappa \le 1, \quad r \ge 1/5$$

$$\Longrightarrow \quad \langle T_b f, g\rangle \le \frac{563371}{10^6} < \frac{40}{71}.$$

There are also the following full-even-block bounds. Put

$$t = r - \frac{389}{1000} + \frac{9}{5}\left(u - \frac{7}{10}\right).$$

Under the common restriction $-1/200 \le t \le 1/50$*, the bounds are*

u	κ	$\langle T_b f, g\rangle$ *at most*
$[.69, .70]$	$[0, 1]$	$551/1000$
$[.70, .72]$	$[0, 1]$	$68/125$
$[.70, .72]$	$[0, 2]$	$451/800$

The first two table caps are below $40/71$*; the last is* 0.56375*. Every finite decimal here denotes its exact rational value.*

Certificate proof. On $0 \le \kappa \le 1$, the accepted Stein row prices give

$$\mathbb{E}|Q| \le \frac{830643632499}{10^{12}} < \rho_1 := \frac{207661}{250000}.$$

The support–source comparison with this cap, $a = 9/5$, and $\lambda = 0$ has directed right side below $1.649920196798 < 1649921/10^6$. Thus every such physical source satisfies

$$r \le \frac{1649921}{10^6} - \frac{9}{5}u.$$

The two accepted rank-one R3 profiles cover the source intervals

$$[1/5, 199/500], \qquad [199/500, 1649921/10^6 - (9/5)u].$$

Their whole-line local proofs use constant reserve $1/50000$ and sixth-degree reserve $1/5000000$. Independent replay accepts all 21873 compact scalar leaves and the exact tails starting at64. Exact $y(r)$ transport and convex endpoint pricing on the two source polygons give respective maxima below 0.563249470254 and 0.563370758117, proving the first cap. The complete replay is recorded in `COMPLETE_RESTRICTED_DOMAIN_REPLAY.json`. The outer extension is recorded in `independent_finer_amplitude_extension.json`.

For the table, the accepted full-even profile uses constant reserve $1/1000$ and sixth-degree reserve $1/10^6$. Its independent whole-line replay checks 56194 partition events, 27636 accepted leaves and 462 exact physical-ellipsoid exclusions, then glues the exact tail beyond64. The six matrix gates and $y(r)$ transport are exact. Let $a_r = b - E_{00} + E_{02}^2/E_{22} < 0$. On each (u, t) rectangle, product-corner interpolation incurs at most

$$(-a_r)\{(9/5)^2(\Delta u)^2 + (\Delta t)^2\}/4.$$

This follows because the mixed term is interpolated exactly and the variance of the affine source is $(9/5)^2 \operatorname{Var}(u) + \operatorname{Var}(t)$; the positive u^2 term only improves the comparison. Source prices are separately convex in κ and the shared shape. All48 vertex prices, including these reserves, have independent directed maxima below 0.550507802111, 0.543476058131, and 0.563687366986. They prove the table caps; the record is `general_r3_source_bands_audit.json`. □

The first conclusion retains both $\kappa \le 1$ and $r \ge 1/5$. The table retains its restrictions on t and κ. Their union is a collection of physical-source regions, and does not establish complete amplitude coverage or a global operator norm bound.

Proposition 26 (Exact transport to a smaller cubic coefficient)**.** *Freeze a valid physical Gaussian profile at $b_0 > 0$, including its whole-line local and scalar majorants. For every $0 < b \le b_0$ all its positive semidefinite gates remain valid. At the same physical parameters $(u, r, \kappa, \text{shape})$, the resulting pairing upper price changes exactly by*

$$(b - b_0)r^2 + c_\beta \left(\frac{1}{b} - \frac{1}{b_0} \right).$$

Here $c_\beta = \beta^2$ for orthonormal longitudinal Hermites and $c_\beta = 6\beta^2$ for raw longitudinal Hermites in the R3 criterion. No new local Gaussian integration is required. In the first case the local potential changes only by its constant $\beta^2(1/b - 1/b_0)$; in the second it is unchanged.

Proof. Each gate at the smaller coefficient is the old gate plus a positive semidefinite diagonal matrix. For example, the raw even gate gains $\mathrm{diag}(0, (b_0 - b)/2)$, and the raw odd gates gain $\mathrm{diag}(0, (b_0 - b)/6)$ or $\mathrm{diag}(b_0 - b, 0)$. The normalized gates behave likewise; the independent gates $E \succeq pp^T$, $E \succeq 0$, $R \succeq 0$ do not change. The rank-one raw gate also weakens.

The only remaining appearances of b in the completed criterion are the source term br^2 and the longitudinal square-completion constant. The norm of the raw third Hermite is squared norm six, accounting for the displayed factor. All source residual and scalar majorant terms are independent of b. In the normalized native criterion the added local constant cancels out of the centered Stein target, so its Stein solution and all derivative inequalities remain the same. Its added constant is simply charged in the outer expectation. This proves the exact price identity. □

Uniform cell pricing and a global coefficient. Suppose an old valid uniform cap U_0 covers a physical source cell $r \in [r_-, r_+]$ with $0 \le r_- \le r_+$. Since $b - b_0 \le 0$, the transported uniform cap

$$U_0 + (b - b_0)r_-^2 + c_\beta(1/b - 1/b_0)$$

is immediately valid. For tighter endpoint repricing, the native source curvature changes to $\xi(b) = \xi(b_0) + b - b_0$, and the full-even R3 curvature changes to $a_r(b) = a_r(b_0) + b - b_0$. Their respective chord reserves must be updated to

$$(-\xi(b))_+(r_+ - r_-)^2/4, \qquad (-a_r(b))_+(r_+ - r_-)^2/4.$$

Every such fixed-profile cell cap is convex in $b > 0$. A fixed selection of profiles covering all physical cells therefore gives a convex maximum of caps. A norm certificate requires one coefficient b valid for the entire cover. Minimizing separately for each sign pair, at $b = \min\{b_0, \sqrt{c_\beta}/r\}$ when $r, c_\beta > 0$, does not prove a bound for a single operator.

Theorem 27 (Complete amplitude branches at one coefficient)**.** *Set $b_* = 7171/10000$. For every pair of odd Gaussian signs f, g, in every finite Gaussian dimension, put $u = \|P_1((f + g)/2)\|_2$. Then*

$$\langle f, T_{b_*} g \rangle \le \begin{cases} 56338/10^5, & 0 \le u \le 1381/2000, \\ 563315250991/10^{12}, & 31/40 \le u \le \sqrt{2/\pi}. \end{cases} \qquad < \frac{40}{71}.$$

Both conclusions include every physical source amplitude, cumulant and cubic tensor shape.

Certificate proof. All the profiles here use the single coefficient b_*. The elementary energy estimate with $\rho = 847/1000$ gives

$$\langle f, T_{b_*} g \rangle \le u^2 + b_* \rho^2 \le \frac{7042043}{12500000} < \frac{40}{71} \quad (0 \le u \le 541/2500),$$

where the displayed cap conservatively uses $b = 18/25$. Four previously replayed complete native frames, transported by Proposition 26, cover $[.14, .42]$, $[.30, .485]$, $[.40, .54]$ and $[.50, .638]$. Their outward caps are respectively below .561498051269, .562840841523, .563045009361 and .563207127166. The intervals overlap each other and the elementary branch.

The exact native-frame union on $[.60, .682]$ consists of 30268 accepted cells and has maximum rational cap $563067382411/10^{12}$. On $[.682, .6905]$, 9661 accepted native cells and 41 cells covered by the transported R3 profile have maximum cap $563379835873/10^{12} < 56338/10^5$. The R3 replacement uses the independently replayed $11470 + 88$ compact leaves and exact tail beyond64, with constant and sixth-degree reserves $1/50000$ and $1/5000000$.

For completeness of these source unions, the support–source proof of (36) applies with the freshly proved $\rho = .847$. For each amplitude subinterval $[u_-, u_+]$, its source grid covers

$$0 \le r \le R_{.847}(u_-),$$

an outward bound on the entire physical source range because $R_{.847}$ is decreasing. The cumulant intervals partition $[0, 90]$. Native prices are affine in the variance shape $s \in [0, 1]$; the R3 coordinate is exactly $z = 2/5 + (3/5)s \in [2/5, 1]$. Every cell has either one complete native bound or one same-coefficient R3 bound. Exact positive source chords and quadratic source maxima, including interior vertices, cover its full continuous interior. Independent price replays check 2240 and 1344 native endpoint prices in these two unions, all selected profile hashes, and the sixteen R3 replacement corners. A separate exact verifier checks all $112 + 42$ amplitude–cumulant groups and every gap-free source partition. This proves the first branch.

The upper native union on $[.775, .797885]$ has 5096 cells and maximum cap $563315250991/10^{12}$; its independent replay checks 1568 endpoint prices and the same exhaustive physical-source partitions. Finally $u \le \sqrt{2/\pi} < .797885$. The complete union records are `native_lower_base_b7171_target_union.json`, `native_R3_b7171_u682_6905_union.json`, and `native_upper_b7171_target_union.json`; their frozen input hashes are checked in `same_b_native_R3_glue_audit.json` and the independent native price replay ledgers. □

The uncovered amplitude range for this theorem alone is $1381/2000 < u < 31/40$. The theorem is an upper bound for the indicated pairings, and does not by itself bound the unrestricted operator norm.

Lemma 28 (Exact pricing on physical source polygons)**.** *Fix an amplitude interval $[u_-, u_+]$, a source interval $[r_-, r_+]$, and a cumulant interval. Suppose that, throughout the cumulant interval and for every allowed shape, proved scalar inequalities imply*

$$a_j u + r \le C_j, \qquad a_j > 0 \quad (1 \le j \le m).$$

Let K be the intersection of these half-planes with the amplitude–source rectangle, after any separately proved coordinate restrictions. For rational inputs its vertices and the maximum on K of

$$F(u, r) = u^2 + L_u u + g r^2 + L_r r + C$$

are computable using rational arithmetic. For a nonempty K, it suffices to check its vertices and, on each edge, the stationary point in the edge interior when the restricted quadratic is concave. This includes segments and singletons. An empty intersection contains no physical source.

For a native certificate interpolated in r, first split K at every source knot. On each resulting piece apply this rule to each of the four cumulant–shape corner bounds, and take the maximum of the resulting bounds. A fixed nonnegative contact multiplier may be included by changing only L_u, L_r, C.

Proof. Every defining half-plane contains every physical source in the nominal rectangle, so intersection and coordinate clipping preserve all physical points. Rational line intersections give all vertices, retaining points on the boundaries. If a full-dimensional polygon had a maximizing point in its interior, sufficiently small horizontal displacements would stay in the polygon. Their two objective values have average $F(u,r)+\delta^2$, contradicting maximality. Thus its maximum lies on its boundary. On the edge $(u,r)=(u_0,r_0)+t(\Delta u,\Delta r)$, $0\le t\le 1$, the quadratic coefficient is $(\Delta u)^2+g(\Delta r)^2$. Endpoints suffice when this coefficient is nonnegative; otherwise also check the unique stationary point if it lies in $(0,1)$. All these operations are rational. The same calculation covers the lower-dimensional cases.

A native source chord is valid only between successive saved knots. Splitting at every knot therefore retains its validity on each closed piece; shared boundaries have both valid endpoint bounds. For each fixed cumulant–shape corner the chord changes only the linear and constant coefficients in r. The already proved corner inequalities and positive interpolation uniformly bound the whole cumulant–shape cell. Taking the maximum over the four corner prices and all source pieces gives a uniform bound. The scalar row chosen at a source endpoint is fixed throughout its cumulant–shape cell; different endpoints may use different globally valid scalar rows.

For a contact with parameters a, B, put $d^2=6B^2$ and let A be its uniform scalar price. The nonnegative multiplier λ adds

$$-\lambda a u-\lambda r+\lambda\left(A+\frac{\beta d}{b}\right)+\frac{\lambda^2 d^2}{4b}.$$

Here β is the normalized native Hermite coefficient; for a raw Hermite R3 profile the product βd is $6\beta_{\mathrm{raw}}B$. The coefficient of u^2 remains one, so the same boundary computation applies. The multiplier and scalar contact are fixed on the whole cell. Selecting any one of several proved upper bounds on each cell preserves validity. □

Uniform source planes. For the accepted even sextic scalar rows, the moment price is affine in shape and convex in cumulant: the sextic coefficient is nonnegative and the coefficient multiplying $\kappa^{3/2}$ is positive throughout the shape interval. Hence the largest of the four endpoint prices is a uniform C_j. A single fixed scalar row is used for each plane over the whole cumulant interval. Taking the smallest uniform price among several such rows is valid; taking separate minima at the four corners would require a separate justification.

Proposition 29 (Base scalar certificate at .713)**.** *Set* $\widehat{b}=713/1000$. *Then*

$$\mathcal{N}(\widehat{b})\le\widehat{B}:=\frac{35245449}{62500000}=.563927184.$$

Consequently the explicit reduced rational matrix sequence has actual limit

$$\lim_{n\to\infty}G(H_n(\widehat{b}))=\frac{1}{\mathcal{N}(\widehat{b})}\ge\frac{62500000}{35245449}=1.7732785869744488\ldots.$$

The rational reciprocal is an unconditional lower bound for $G_{\mathbb{R}}$. *The scalar certificate does not assert* $\mathcal{N}(\widehat{b})=\widehat{B}$.

Certificate proof. All branches use the same coefficient $\widehat{b}$. The independently verified lower-amplitude ledger covers $u\in[0,1381/2000]$ with bound $563295214999/10^{12}<\widehat{B}$. It combines the low-energy branch, four transported regional frames and two complete native–R3 source unions. The exact transfer from $\bar{b}=7171/10000$ is

$$(\widehat{b}-\bar{b})r^2+\beta_{\mathrm{norm}}^2(1/\widehat{b}-1/\bar{b}),$$

where $\beta^2_{\mathrm{norm}} = 6\beta^2_{\mathrm{raw}}$ for an R3 profile. The selected source unions retain the nonpositive first term; the spectral gates remain valid at the smaller coefficient. The lower ledger checks all 39929 native cells and 41 R3 cells in these two unions, along with the four earlier regional frames.

The original upper cover has 25 amplitude bands on $[1381/2000, 159577/200000]$ and 56 cumulant bands on $[0, 90]$. The endpoint $159577/200000$ exceeds the maximum physical amplitude $\sqrt{2/\pi}$. Every one of its 1400 groups covers every physical source from zero to an outward $R_{.847}(u_-)$ endpoint, with all shapes included. Native shape $s \in [0,1]$ corresponds exactly to $z = 2/5 + (3/5)s$ for the R3 prices. Directed checks verify all 25 original source-grid upper endpoints and 250 inverse clips. Nineteen initial R3 profiles give 1064 cumulant-band constants, checked at all 4256 cumulant–shape corners. Seven native tables have 11648 independently checked baseline endpoint prices, followed by the exact transfer above. Their source quadratics also gain $\widehat{b} - \bar{b}$.

Each cumulant band has ten proved uniform positive-head source planes. The 560 planes and their moment corner prices are independently verified. Of the original 255360 source cells, 253282 retain upper prices and 2078 are exactly empty. The producer-free exact replay checks every original interval, selected polygon or containing rectangle, native radial knot and concave edge stationary point, using Lemma 28. It reconstructs all selected contact and ordinary prices and accounts for every original cell exactly once.

Three additional independently accepted whole-line profiles strengthen 1069 prices on the identical cover. An exact amplitude refinement then replaces 2814 difficult parent cells by 27340 child intervals: 20749 are nonempty and 6591 are exactly empty. The other 250468 price cells remain unchanged. Each child inherits its parent's proved source restrictions; its clipped polygon and selected price are checked independently. The old parent bound is retained as an option. Five further accepted whole-line profiles then strengthen prices on this same refined cover in three successive monotone stages, changing 2397, 27889 and 2428 prices respectively. Their 224, 448 and 448 new cumulant–shape corner prices are independently recomputed. The resulting 27-profile cover has 271217 price cells and 8669 exact empty cells.

Finally, introduce a 68-point cumulant grid while retaining every previous bound. Only 364 difficult parents are split, giving an exact partition into 1794 cumulant children. Of these, 1726 retain prices and 68 are exactly empty; the remaining 270853 price cells are unchanged. Seven native endpoint tables on this finer grid have 13936 independently checked baseline corner prices, transported exactly to $\widehat{b}$. Fourteen new cumulant bands use ten fixed positive-head planes each, chosen from the already proved 60-row analytic bank. All 140 planes are independently checked at their 560 moment corners. Their intersections contain every physical child point. An exact replay verifies every child cumulant partition and complete clipped polygon. It reconstructs 1110 newly selected native prices, splitting every radial knot and including every concave edge stationary point; the other 616 nonempty children retain their valid parent bounds. No new R3 constant is needed in this last refinement.

Two final independently accepted whole-line profiles are added on this identical cumulant-refined cover. They strengthen 691 and 343 prices respectively, while retaining every prior bound. Each new profile has 280 independently recomputed cumulant–shape corner prices on the mixed collection of original and refined bands. An exact replay reconstructs every changed polygonal or rectangular price and verifies that all other cell records and all empty cells are unchanged.

The final complete upper cover thus contains 272579 price cells and 8737 exact empty cells. Its maximum recorded price is

$$\frac{4511417459}{8 \cdot 10^9} < \widehat{B}.$$

The frozen source is `global_upper_b713_native_kappa68_twentyninebank_envelope.json.gz`, with SHA-256 `ea763375a19e63a49174bcf18975c948ab188f993edfb471014cbb2bfec04d2d`. The

final independent reconstruction is `fullupper_b713_native_kappa68_twentyninebank_monotone_audit.json`; it binds the complete chain of accepted parent covers and replay hashes.

These upper and lower branches therefore bound every odd sign pairing in every finite Gaussian dimension. Negating one sign gives the absolute norm bound. The established Gaussian-to-matrix convergence and exact ratio-preserving reductions give the actual matrix limit and lower bound. □

Theorem 30 (Complete transported scalar certificate)**.** *Set* $\widehat{b} = 711/1000$*. Then*

$$\mathcal{N}(\widehat{b}) \le \widehat{B} := \frac{112782853}{2 \cdot 10^8} = .563914265.$$

Consequently the explicit reduced rational matrix sequence has actual limit

$$\lim_{n\to\infty} G(H_n(\widehat{b})) = \frac{1}{\mathcal{N}(\widehat{b})} \ge \frac{2 \cdot 10^8}{112782853} = 1.7733192119195636\ldots.$$

The rational reciprocal is an unconditional lower bound for $G_{\mathbb{R}}$*; the norm need not equal its certified cap.*

Coefficient transport. Retain the complete upper physical partition and every selected frozen criterion from Proposition 29, at $b_0 = 713/1000$. Amplitude, source, cumulant, shape and all physical source restrictions are independent of the operator coefficient. All 8737 empty cells therefore remain empty.

For an ordinary frozen criterion let β_i be its normalized longitudinal coefficient: it is the recorded native coefficient, or $\sqrt{6}$ times the recorded raw R3 coefficient. Its pointwise price changes by

$$(\widehat{b} - b_0)r^2 + \beta_i^2(1/\widehat{b} - 1/b_0).$$

For a fixed head-contact criterion, with multiplier $\lambda_i \ge 0$ and head $a_c Z + B_c H_3(Z)$, put $d_c = \sqrt{6}B_c$. Completing the same longitudinal square gives instead

$$(\widehat{b} - b_0)r^2 + (\beta_i + \lambda_i d_c/2)^2(1/\widehat{b} - 1/b_0).$$

Thus the ordinary case is included by taking $\lambda_i = 0$. These identities follow by grouping the longitudinal cost β_i^2/b, its contact cross term $\lambda_i\beta_i d_c/b$ and the contact square cost $\lambda_i^2 d_c^2/(4b)$. All other frozen prices are unchanged. All spectral matrix gates are checked at the smaller coefficient.

Let a_i be the accepted old upper price and $r_{i,-}$ its source cell's lower endpoint. Since $\widehat{b} - b_0 < 0$ and every physical point in that cell satisfies $r \ge r_{i,-} \ge 0$, a valid new uniform price is

$$a_i + (\widehat{b} - b_0)r_{i,-}^2 + (\beta_i + \lambda_i d_c/2)^2(1/\widehat{b} - 1/b_0).$$

An inherited parent bound is treated in precisely the same way, using its original frozen criterion and multiplier; it already bounds every physical child point. The coefficient $\widehat{b}$ is fixed throughout all cells and is not chosen separately for different sign pairs.

An independent producer-free replay reconstructs all 43 exact profile bindings, verifies the target spectral gates and evaluates these shifts on every one of the 272579 price cells. The final assigned mosaic has 272557 ordinary cells and 22 R3 contact cells. Every one of their transport shifts is rational and is computed exactly. The maximum rounded shifted price is

$$\frac{563914263671}{10^{12}} < \widehat{B}.$$

The source is the hash-bound `assigned_twenty-nine_coefficient_scan.json`; the exhaustive independent cell replay is `assigned_b711_shifted_cells_audit.json`. The separately verified complete lower-amplitude branch at the same coefficient, $u \in [0, 1381/2000]$, has upper bound $563307766601/10^{12} < \widehat{B}$. It includes the elementary branch, four earlier regional frames and both complete lower source unions. Together the transported upper and lower covers contain every physical odd sign pair in every finite Gaussian dimension. Negating one sign gives the absolute norm bound. Gaussian-to-matrix convergence and the exact ratio-preserving reductions give the stated actual matrix limit. □

11 Consequences for search and reflection representations

The sequence lies in the full absolute doubly stochastic class. Formula (15) now performs that conversion. Evaluate the copies on their smaller seed, merge proportional rows and columns with their total absolute weights, and quotient signed row/column permutations.

At fixed order, decompose $|B|$ into a convex combination of permutation matrices and restore the fixed entry signs, giving signed permutations with simplex weights and no cancellation. A signed left permutation fixes the first term to the identity. Disconnected support ratios are C-weighted component averages, so keep a best component. A balanced cycle-sign support gauges to a nonnegative matrix and has ratio one. For two signed permutations with weights $t, 1-t$, each nontrivial component is an alternating cycle on $k \geq 2$ rows and k columns. A frustrated cycle has $C = k - 2\min(t, 1-t)$ and

$$V \leq k\sqrt{1 - 4t(1-t)\sin^2(\pi/(2k))}.$$

To see the latter bound, gauge its matrix to $tI + (1-t)S$, where the signed cyclic permutation has $S^k = -I$, and use $V \leq k\|\cdot\|_{\mathrm{op}}$. For $0 \leq t \leq 1/2$ the squared ratio is increasing, since $k\sin^2(\pi/(2k)) \leq 1$. Its maximum is at most $k\cos(\pi/(2k))/(k-1)$, which is $\sqrt{2}$ at $k = 2$ and at most $3/2$ for $k \geq 3$. Thus a search near 1.775 needs frustrated cycles and at least three permutation terms; this does not assert that three suffice.

A weighted-reflection representative is also explicit. Since $|H_n|$ is doubly stochastic, $\|H_n\|_{\mathrm{op}} \leq 1$. If its order is M_n, put

$$U_n = \begin{pmatrix} H_n & (I - H_n^2)^{1/2} \\ (I - H_n^2)^{1/2} & -H_n \end{pmatrix}, \quad P_n = (I - U_n)/2, \quad D_n = \mathrm{diag}((2/M_n)I_{M_n}, 0).$$

Then $P_n = P_n^T = P_n^2$, its diagonal is $1/2$, and

$$D_n - 2D_n^{1/2}P_nD_n^{1/2} = (2/M_n)H_n \oplus 0.$$

Its ratio is unchanged. Zero auxiliary weights are essential to this universal representation. Strictly equal positive weights on every coordinate would give an orthogonal matrix, whose ratio is at most $\pi/2$, and would wrongly exclude the target. Indeed, for an orthogonal matrix O of order m, $V(O) \leq m$. Gaussian signs of Z_i and $(O^TZ)_j$ give

$$C(O) \geq \frac{2}{\pi}\sum_{ij} O_{ij}\arcsin(O_{ij}) \geq \frac{2}{\pi}\sum_{ij} O_{ij}^2 = \frac{2m}{\pi},$$

so $G(O) \leq \pi/2$.

References

[1] B. Peng, *A computer-assisted upper bound of 1.7813 for the real Grothendieck constant*, arXiv:2609.20074v1, September 17, 2026.

[2] R. Saha, A. Li, A. Xue, S. Chaudhuri, A. Klivans, P. K. Kothari and R. Meka, *New Lower and Upper Bounds for the Grothendieck Constant*, arXiv:2608.11158v2, 2026.

[3] Retained project report, *Real Grothendieck continuation*, September 30, 2026, Theorem 3 and its inherited analytic premises.

[4] Retained project report, *Matrix invariance and stochastic normalization*, September 30, 2026; independently reconstructed reductions in this continuation.

[5] Recovered matrix-mining sign certificates and *Baseline and transfer review*, October 1, 2026, included in the source package. The fixed-coefficient sign witnesses have fresh directed numerical replays; the historical global norm cover remains a separate premise.

[6] *Grothendieck 177128*, preceding delivered note and source package, October 1, 2026. Its endpoint depends on the cited Song–Yue uniform norm theorem and the stated nonlinear-cover scope.

[7] S. Heilman, C. Jones and G. Malavolta, *Sharper Bounds for the Complex Grothendieck Constant*, arXiv:2609.25462v1 (2026), equation (8.12). The tensor identities used here are also freshly checked by connected Wick diagrams.

A Exact tensor comparison

We give the finite acceptance rules for (27). Set $x = 1 - y$ and $z = 1 - a$. Each of 84 leaves specifies a closed rational rectangle in $[0,1]^2$, one of the upper bounds (25)–(26), and rational tangent parameters. For $t > 0$ the inequalities

$$\sqrt{X} \le (X + t^2)/(2t), \qquad \sqrt{XY} \le (tX + Y/t)/2$$

replace square roots by polynomial upper bounds. For a cube $X^{3/2}$ multiply the first inequality by $X \ge 0$. Choose rational $y_0 \in [0,1]$ and $l_0 > 0$ with $l_0^2 \ge 1 + 3y_0^2/2$. Cauchy–Schwarz gives

$$\sqrt{d} \ge L(y) = (1 + 3y_0 y/2)/l_0.$$

The bracket $[(K + 422/15)d - 211/3]$ is positive, since its minimum is $K - 211/5 > 0$. It is enough to show that the selected polynomial upper bound for $\mathcal{M}$ or $\mathcal{J}$ is at most this bracket times $L(y)$.

The dedicated equality-corner rectangle is $x, z \in [0, 2^{-19}]$. It uses $(p, q) = (3/4, -1)$ and $L(y) = (2 + 3y)/\sqrt{10}$. Here $(1 - y^2)^{3/2} \le (1 - y^2)/512$; tangent parameters $7/4$ and 1 handle the remaining roots. This retains equality at the corner.

Each difference polynomial has degree at most $(4, 4)$ over $\mathbb{Q}(\sqrt{10})$. After substitution of the affine rectangle coordinates, let c_{kl} be its power coefficients. Its Bernstein coefficients are

$$b_{ij} = \sum_{k \le i,\, l \le j} c_{kl} \frac{\binom{i}{k}}{\binom{4}{k}} \frac{\binom{j}{l}}{\binom{4}{l}}, \qquad 0 \le i, j \le 4.$$

Every coefficient is nonnegative. The sign of $r + s\sqrt{10}$ is decided by rational sign cases and comparisons of r^2 with $10s^2$. Since the Bernstein basis is nonnegative and sums to one, the 25 checks prove the entire rectangle. The leaf counts are 21, 36 and 7 for the first three $\mathcal{J}$ bounds, 1 and 18 for the two Gram branches, and 1 for the dedicated corner: 84 rectangles and 2100 coefficients.

The rectangles lie within the square, have pairwise disjoint interiors, and have exact total area one. Their finite closed union covers the square: an omitted point would have a relatively open neighborhood of positive area. This checks every boundary as well as the interior. Two independent programs reconstruct the polynomials from the formulas. Neither imports the producer's coefficient lists or the older 99-rectangle data.

B Whole-line scalar certificates

All parameters of (32) and (31) are frozen in exact rational row files. The rows and complete cumulant cover are shown below; displayed prices are rounded upward, while acceptance uses directed rational enclosures.

Row	Cumulant interval	Corner price upper
0	$[0, 39/100]$	.825942224105
2	$[39/100, 39/25]$	.839598147439
3	$[39/25, 4]$	.847061071421
4	$[4, 34/5]$	.847358899934
$k8$	$[34/5, 48/5]$	.848520760642
5	$[48/5, 76/5]$	.848507934475
6	$[76/5, 28]$	.846999761374
7	$[28, 90]$	.838404593017

For each row, 512-bit outward Arb arithmetic and fourth-order Taylor bounds verify the entire compact interval $[0, 64]$ on an adaptive rational partition. Mills' ratio is enclosed at the endpoints, and $R' = xR - 1$ generates the needed Taylor coefficients. For $x \geq 64$, $0 \leq 1 - xR(x) \leq x^{-2}$ and convexity of the square reduce the comparison to two rational polynomial tests. Clearing the positive denominators and substituting $x = 64 + t$ gives nonnegative coefficients and positive constants, proving the complete unbounded tail. Evenness supplies the negative half-line.

The independent verifier uses direct Chebyshev recurrence, 768-bit arithmetic and sixth-order Taylor bounds on $[0, 80]$, followed by a separately reconstructed exact tail at 80. It also independently assembles all 32 prices in (33). A further checker reconstructs the 16 tail comparisons and 32 corner prices without importing either scalar implementation. Complete comparisons, rather than sampled maxima, are the acceptance criterion.

C Polynomial certificate for the cubic source gate

The third numerator in Proposition 10 is explicitly

$$\begin{aligned} P_3 = {} & A^6/8 + A^5B(3w/2 + 21/4) \\ & + A^4B^2(57w^2/8 + 195w/4 + 84) \\ & + A^3B^3(17w^3 + 339w^2/2 + 540w + 504) \\ & + A^2B^4(171w^4/8 + 549w^3/2 + 1194w^2 + 1728w) \\ & + AB^5(27w^5/2 + 837w^4/4 + 1116w^3 + 1980w^2) \\ & + B^6(27w^6/8 + 243w^5/4 + 378w^4 + 756w^3). \end{aligned}$$

Every coefficient is nonnegative for $A > 0$, $B, w \geq 0$. Thus this finite identity establishes the derivative sign on the entire half-line; no numerical Gaussian quadrature is used. As an exact example, take $A = 97/40$, $B = 1/6$. Then

$$a = 117/40, \quad d = \sqrt{6}/6, \quad (q_0, q_2, q_4, q_6) = \left(\frac{331}{120}, \frac{20}{97}, -\frac{1096000}{265587843}, \frac{106274560000}{727184560303017}\right).$$

At $\kappa = 53/100$, $u = 151/200$, $v = -1/4$, the two variance endpoints give the directed source caps

$$r < 252844293/10^9 \quad (V = 7\kappa/45), \qquad r < 250945617/10^9 \quad (V = 2\kappa/9).$$

The same affine shape interpolation applies between them. These conditional examples illustrate the gate; $v = -1/4$ is not assumed for arbitrary optimizing signs. Independent exact polynomial and directed price implementations verify the recurrence, integration constants and two rational caps.

D Complete continuous regional certificates

For $T = P_1 - (18/25)P_3$, odd signs f, g, and $u = \|P_1((f+g)/2)\|_2$, the following complete certificates give the displayed bounds on $\langle f, Tg\rangle$:

amplitude interval	local boxes	scalar rows	rational upper cap
$[7/50, 21/50]$	63681	24	$561523/10^6$
$[3/10, 97/200]$	61829	24	$562857/10^6$
$[2/5, 27/50]$	58841	24	$563039/10^6$
$[1/2, 319/500]$	151929	24	$563266/10^6$
$[3/5, 331/500]$	153593	24	$563302/10^6$
$[331/500, 341/500]$	627497	24	$563367/10^6$
$[77/100, 159577/200000]$	53541	24	$563334/10^6$

Every cap is strictly below $40/71$. The respective directed totals are below 0.561522410, 0.562856131, 0.563038808, 0.563265432, 0.563301723, 0.563366577, and 0.563333793. The first profile uses the universal transverse bound $r \leq 849/1000$ and therefore remains valid throughout its extended amplitude interval. The other extensions are checked against the outward source endpoint. The outer polynomial $u^2 - 2\ell u + C$ is convex, so rational endpoint checks cover each entire interval.

Each certificate has an optimizer-free, directed-arithmetic replay of all physical moment boxes, scalar compact leaves, infinite scalar tails, the five matrix gates, and the continuous cumulant, shape, and source interpolations. Infinite tails explicitly begin beyond every target and regression hinge. Local height and scalar coefficient repairs are fully priced. These are regional norm bounds; amplitudes outside the stated coverage still require certification before a new unrestricted norm cap or Grothendieck lower endpoint follows. The primary union is $[.14, .682] \cup [.77, \sqrt{2/\pi}]$. The separately proved low-energy branch covers $[0, .14]$. Thus these earlier regional certificates alone leave the target amplitude interval $(.682, .77)$ unclosed; the later complete cover is treated separately. The earlier, tighter bound $9/16$ on $[.79, \sqrt{2/\pi}]$ and its original portable certificate are retained as supplementary results.

E Original-degree accounting for the sign witness

In this appendix $H_j = \mathrm{He}_j/\sqrt{j!}$ denotes the orthonormal longitudinal Hermite polynomial. The central limit theorem for Y_N alone does not justify assigning it ordinary Hermite degree one. Each summand has original degree three. Here we verify all projections used in (19).

First replace the sign by a bounded smooth approximation with bounded derivatives. Expand in one transverse summand $H_3(X_i)/\sqrt{N}$. Orthogonality and parity give

$$\mathbb{E}[H_1 H_3] = \mathbb{E}[H_1 H_3^2] = \mathbb{E}[H_2 H_3] = 0.$$

Taylor expansion therefore makes the $H_1(X_i)$ coefficient $O(N^{-3/2})$ and the $H_2(X_i)$ coefficient $O(N^{-1})$. The $H_3(X_i)$ coefficient is $N^{-1/2}\mathbb{E}\partial_y f + O(N^{-3/2})$. Expand separately in distinct transverse coordinates for mixed terms. The complete list of original-degree-one and degree-three basis types is

Basis type	Coefficient order	Summed squared energy
$H_1(Z)$	$\to m_1$	retained
$H_1(X_i)$	$O(N^{-3/2})$	$O(N^{-2})$
$H_3(Z)$	$\to m_3$	retained
$H_2(Z)H_1(X_i)$	$O(N^{-3/2})$	$O(N^{-2})$
$H_1(Z)H_2(X_i)$	$O(N^{-1})$	$O(N^{-1})$
$H_1(Z)H_1(X_i)H_1(X_j)$	$O(N^{-3})$	$O(N^{-4})$
$H_3(X_i)$	$N^{-1/2}m_Y + o(N^{-1/2})$	$\to m_Y^2$
$H_2(X_i)H_1(X_j)$	$O(N^{-5/2})$	$O(N^{-3})$
$H_1(X_i)H_1(X_j)H_1(X_k)$	$O(N^{-9/2})$	$O(N^{-6})$

The indices i, j, k are distinct. For a fixed smooth approximation the orders are uniform in these indices; the last column counts every term of the displayed type. The ordinary central limit theorem and normal integration by parts identify the three retained moments. Reflection of Y_N preserves m_1, m_3 and changes m_Y to $-m_Y$. Thus the degree-one and degree-three cross-pairings tend respectively to m_1^2 and $m_3^2 - m_Y^2$.

Finally $q'(z) \geq 511/200 > 0$. Conditional on Y_N, the normal density and this derivative bound give a small-band estimate

$$\Pr(|Y_N + q(Z)| \leq \varepsilon) \leq \frac{2\varepsilon}{(511/200)\sqrt{2\pi}},$$

uniformly in N. The same holds for the reflected witness. Consequently the smooth approximation can be removed uniformly in L_2; orthogonal projection is an L_2 contraction. This completes the limiting degree calculation. The exact witness parameters, directed compact integration and infinite-tail bounds are supplied in `reduced_search/operator/cubic720/README.md` and the adjacent `cubic720_norm_witness.json`.

F Reproduction, retained premises and proof status

The commands and numerical tables in this appendix describe the retained October 1 checkpoints. The completed continuation is stated in Appendix L, where the complete same-coefficient certificate proves $G_{\mathbb{R}} \geq 1.775$ and the full proof gives the three-significant-digit conclusion 1.78.

The source archive contains the exact row inputs, tensor rectangles, complete-halfline checkers, formal Wick enumeration, implicit matrix construction, independent audits, and directed MPFR numerator and sign witness calculations. The main checks can be run from the archive root:

```
python reduced_search/energy/633/certify_wick_identity.py
python reduced_search/energy/633/discover_cover.py --replay
python reduced_search/energy/633/independent_replay.py
python reduced_search/energy/certify_stein_cubic_L1.py
python reduced_search/energy/audit_stein_L1_independent.py
python tonight/moments/certify_cubic_head_polynomial.py
python tonight/theory/certify_numerator_rate.py
python tonight/theory/certify_retained_finite_ratio.py
python tonight/theory/certify_saha_benchmark.py
python tonight/theory/certify_published_upper_rounding.py
python tonight/operator/cubic_L1_refinement/refit847/verify_cover.py
python tonight/audit/refit847/replay_complete_refit_cover.py
python tonight/regional_certificates/u014_042/code/verify_region.py
python tonight/regional_certificates/u030_0485/code/verify_region.py
python tonight/regional_certificates/u040_054/code/verify_region.py
python tonight/regional_certificates/u050_0638/code/verify_region.py
python tonight/regional_certificates/u060_0662/code/verify_region.py
python tonight/regional_certificates/u0662_0682/code/verify_region.py
python tonight/regional_certificates/u077_c/code/verify_region.py
python tonight/operator/projection/replay_R3_cell/verify_all.py
python tonight/operator/projection/replay_accepted_R3/verify_all.py --quick
python tonight/operator/preferred_global/verify_bundle.py
python reduced_search/operator/cubic7171/certify_cubic7171_norm_witness.py
python tonight/audit/replay_cubic7171_witness.py
python tonight/audit/replay_selected_cubic_witnesses.py
```

Python, python-flint and the system MPFR library suffice for these accepted checks; discovery programs have additional numerical dependencies. Files are bound by a SHA-256 manifest. Floating optimization identifies candidate inputs and is kept separate from proof outputs.

For $b = .72$, directed evaluations of (14) give

n	Certified vector numerator lower bound
32	.999159371292
128	.999947506710
1024	.999999179839

These bounds are unconditional numerator estimates. A denominator bound valid for all sign pairs in every dimension is still required before (17) gives the target lower endpoint.

The global cap in Theorem 30, inserted into (17) at $b = 711/1000$, gives

n	Unconditional lower bound for $G(H_n(.711))$
32	1.771831610393
128	1.773226593281
1024	1.773317765118

These displayed lower bounds converge to 200000000/112782853. The actual ratio converges to $1/\mathcal{N}(.711)$, which may be larger. All finite rounding losses, including the total entry error n^{-4}, are charged by the directed checker `evaluate_finite_ratio.py`.

For comparison, the retained .699 denominator premise already gives explicit finite members of the reduced sequence. Directed evaluation of (17), with that premise stated separately, yields

n	Conditional lower bound for $G(H_n(.699))$
32	1.771523050434
128	1.772908047308
1024	1.772998563604

These conditional finite lower bounds converge quantitatively to 1.773; the actual ratio limit remains $1/\mathcal{N}(.699)$. These numbers use the earlier scalar norm premise and do not reprove it. The very large matrices are specified implicitly by their analytic seed and exact cloning maps.

The inherited project estimate (2) remains an explicit premise. Its 55-interval cover and inherited integrations are not reconstructed by the present archive. The exact reductions, asymptotic sequence, coupled cumulant theorem, cubic absolute-moment theorem, and support–source propositions above have their own fresh proofs and do not depend on that missing reconstruction.

The conditional project lower endpoint is 1.773, at distance .002 from 1.775. The later 1.7712846852... family was weaker and does not replace it. The fixed .699 construction has a directed ceiling 1.77440101857...; the fresh .72 ceiling is 1.77543615103..., and the fresh .7171 ceiling is 1.775462496412. These ceilings bound the reciprocal-norm methods, not the unrestricted Grothendieck constant. The regional amplitude caps and the complete same-coefficient global envelope prove the fresh October 1 unconditional lower endpoint 1.7733192119.... The refined complete certificate in Section I improves that checkpoint using independently accepted source prices and a complete refined cover. The final certificate in Appendix L reaches 1.775 and completes the three-significant-digit determination.

G Finite odd multipliers with an identity tail

Let $P_j^{(d)}$ be the orthogonal projection onto total Gaussian Hermite degree j, and let $P_{\mathrm{odd}}^{(d)}$ be antipodal odd projection. Fix computable real coefficients $\tau, c_0, \dots, c_K$ and put

$$T^{(d)} = \tau P_{\mathrm{odd}}^{(d)} + \sum_{k=0}^{K} c_k P_{2k+1}^{(d)}, \qquad \lambda_{2k+1} = \tau + c_k \ (k \le K).$$

Its remaining odd eigenvalues are τ. Suppose

$$\tau + c_0 = 1, \qquad |\tau| \le 1, \qquad |\tau + c_k| \le 1 \quad (1 \le k \le K). \tag{58}$$

Define

$$N(T) = \sup_{d \ge 1} \sup_{|f|,|g| \le 1} \langle f, T^{(d)} g \rangle.$$

The absolute value may equivalently be inserted in this definition. Odd signs suffice: odd projection preserves the unit ball of L_∞, and a bilinear functional over bounded odd functions has the same supremum as over odd signs.

Proposition 31 (Exact radial Hermite energies)**.** *For $X \sim \gamma_d$, write $R = \|X\|$, $V = X/R$, and*

$$a_d = \frac{(\mathbb{E}R)^2}{d} = \frac{2\Gamma((d+1)/2)^2}{d\Gamma(d/2)^2}, \qquad \alpha = d/2.$$

With the generalized Laguerre polynomial L_k^α, one has

$$P_{2k+1}^{(d)} V_i = \frac{\mathbb{E}R}{d} \frac{(1/2)_k}{(\alpha+1)_k} X_i L_k^\alpha(R^2/2), \tag{59}$$

$$e_{2k+1}(d) := \sum_i \|P^{(d)}_{2k+1} V_i\|_2^2 = a_d \frac{(1/2)_k^2}{k!(\alpha+1)_k}. \tag{60}$$

In particular

$$e_1 = a_d, \quad e_3 = \frac{a_d}{2(d+2)}, \quad e_5 = \frac{9a_d}{8(d+2)(d+4)}, \quad e_7 = \frac{75a_d}{16(d+2)(d+4)(d+6)}.$$

The exact natural-vector objective is

$$q_d(T) = \tau + \sum_{k=0}^{K} c_k e_{2k+1}(d) \longrightarrow 1. \tag{61}$$

Proof. Gaussian projection commutes with orthogonal rotations. Thus the projection of the equivariant vector field V onto degree $2k+1$ is a constant times $X_i L_k^\alpha(R^2/2)$. Indeed an equivariant vector polynomial is X_i times a radial polynomial, and orthogonality to the lower odd degrees is precisely Laguerre orthogonality with weight $t^\alpha e^{-t}$. Put $t = R^2/2$. Radial integration and Laguerre orthogonality give

$$\mathbb{E}[X_i^2 L_k^\alpha(t)^2] = \frac{(\alpha+1)_k}{k!}, \qquad \mathbb{E}[V_i X_i L_k^\alpha(t)] = \frac{\mathbb{E}R}{d}\frac{(1/2)_k}{k!}.$$

For the second identity, use

$$\int_0^\infty t^{\alpha-1/2} e^{-t} L_k^\alpha(t)\, dt = \Gamma(\alpha+1/2)\frac{(1/2)_k}{k!},$$

which follows by integrating the generating function $(1-z)^{-\alpha-1}\exp[-tz/(1-z)]$ and comparing coefficients. Taking the ratio proves (59); its squared norm proves (60). Parseval gives $\sum_{k\geq 0} e_{2k+1}(d) = 1$. Finally $a_d \to 1$ and every fixed $e_{2k+1}(d)$ with $k \geq 1$ tends to zero. Since $\tau + c_0 = 1$, (61) follows. □

Proposition 32 (An explicit reduced matrix sequence)**.** *For $n \geq 2$, set $d = n^2$. Partition each Gaussian coordinate into intervals of mesh n^{-2} from $-n$ to n, together with the two exterior intervals. Retain one cell C from each antipodal pair, using the cells with first coordinate negative. Set*

$$w_C = 2\gamma_d(C), \qquad D_w = \mathrm{diag}(w_C), \qquad F_j[C,\xi] = 2\int_C H_\xi\, d\gamma_d \quad (|\xi| = j),$$

where H_ξ is the normalized product Hermite polynomial. Define

$$A_n(T) = \tau D_w + \sum_{k=0}^{K} c_k F_{2k+1} F_{2k+1}^T. \tag{62}$$

Then

$$C(A_n(T)) \to N(T), \qquad V(A_n(T)) \to 1, \qquad G(A_n(T)) \to \frac{1}{N(T)}. \tag{63}$$

Choose dyadic rational approximants with total entrywise error at most $\eta_n = n^{-4}$. Clear their denominators, delete zero rows and columns, and apply the exact absolute row/column cloning reduction, the symmetric off-diagonal dilation, and the signed balanced copy. The resulting rational matrices $H_n(T)$ are symmetric, zero-diagonal, have zero row and column sums, and have $|H_n(T)|$ doubly stochastic. Their ratios have the same limit $1/N(T)$.

Proof. Odd extension of the cell values turns the finite bilinear form (62) into $\langle f, T^{(d)} g\rangle$. The diagonal term is exact because for odd cell functions

$$\langle f, P^{(d)}_{\mathrm{odd}} g\rangle = \sum_C w_C f_C g_C.$$

The remaining terms are the exact Hermite projection pairings. Thus $C(A_n(T)) \le N(T)$. Every pair of bounded odd functions in a fixed dimension can be approximated in L_2 by conditional averages on the first coordinates of these grids as $n \to \infty$. Continuous compactly supported approximation proves this even though the grids are not nested. The other coordinates integrate out, and $T^{(d)}$ is consistent under such cylindrical embeddings. Since its L_2 norm is one, the pairings converge. Finite cube maxima occur at sign vertices, proving $\liminf C(A_n(T)) \ge N(T)$.

For vector-valued odd cell functions of pointwise norm one, the same L_2 operator bound and Cauchy–Schwarz give $V(A_n(T)) \le 1$. Use the unit vector at the clipped cell midpoint as the trial field. The quantitative estimate below proves that its objective tends to $q_d(T)$ and hence to one. This proves the vector limit. Positivity is explicit: for $s = \operatorname{sgn} X_1$, first-chaos energy is $2/\pi$ and all remaining odd eigenvalues are at least -1, whence $N(T) \ge \langle s, Ts\rangle \ge 4/\pi - 1 > 0$.

Both finite norms change by at most total entrywise approximation error. Consequently rational approximation preserves both limits. For completeness, if Z is the cleared integer seed, put $r_i = \sum_j |Z_{ij}|$, $s_j = \sum_i |Z_{ij}|$, and replace its entry by $Z_{ij}/(r_i s_j)$ on each of the r_i row and s_j column copies. Both norms are unchanged, because every original row and column has total absolute copying weight one; the absolute matrix is doubly stochastic. Off-diagonal symmetric dilation multiplies both norms by two. Tensoring with $\frac{1}{2}\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$ multiplies both by two again and imposes zero sums. All ratios are therefore preserved exactly. □

Proposition 33 (The general asymptotic vector value)**.** *Drop* (58) *and assume only that T is nonzero. Write*

$$M = \max\bigl(|\tau|, |\tau + c_0|, \dots, |\tau + c_K|\bigr).$$

Choose nonzero rational approximants, or discard the finitely many initial zero approximants before applying the reductions. This is possible since $C(A_n(T)) \to N(T) > 0$. The seed (62) *and these rational reductions satisfy*

$$C(A_n(T)) \to N(T), \qquad V(A_n(T)) \to M, \qquad G(H_n(T)) \to M/N(T).$$

For every fixed odd Hermite degree j and $d \ge j$, the square-free tensor trial field used in the proof has norm-square variance

$$\epsilon^2_{d,j} = \sum_{r=1}^{j} \frac{\binom{j}{r}\binom{d-j}{j-r}}{\binom{d}{j}}(3^r - 1) \le \frac{j(3^j - 1)}{d}.$$

Terms with an impossible intersection size have binomial coefficient zero. Its unit normalization attains $|\lambda_j|$ within $2M\epsilon_{d,j}$ in continuum vector objective.

Proof. Let $m = \binom{d}{j}$ and define the vector with components indexed by j-subsets S of $\{1, \dots, d\}$,

$$Y_S(X) = m^{-1/2} \prod_{i \in S} X_i, \qquad U = Y/\|Y\|.$$

The exceptional zero set has Gaussian measure zero. Every component has pure degree j, and $\mathbb{E}\|Y\|^2 = 1$. For two subsets S, S', their norm-square moment is $3^{|S \cap S'|}$. Counting intersections

proves the variance formula. If R is their intersection size, $\mathbb{E}R = j^2/d$ and $3^r - 1 \le r(3^j - 1)/j$ for $0 \le r \le j$, giving its upper bound. The elementary inequality $(\sqrt{x} - 1)^2 \le (x - 1)^2$ yields

$$\|U - Y\|_{L_2}^2 = \mathbb{E}(\|Y\| - 1)^2 \le \epsilon_{d,j}^2.$$

Thus $|\langle U, TU\rangle - \lambda_j| \le 2M\epsilon_{d,j}$. If $\lambda_j < 0$, use $-U$ on the right. Any selected nonzero odd eigenvalue is covered; for the tail eigenvalue choose an odd $j > 2K + 1$. This proves that the continuum supremum of the vector objective over all dimensions is M.

For each fixed trial dimension, conditional cell averages of U on the first coordinates converge in L_2 as $n \to \infty$. They have pointwise norm at most one; completing left and right ball vectors in separate orthogonal coordinates gives admissible finite unit vectors with the same cross pairings. Therefore the seed vector norms have $\liminf \ge M$. The L_2 norm of T gives the reverse bound. The scalar convergence and rational reductions are the same as in the preceding proof, with operator norm M replacing one. A nonzero multiplier has $N(T) > 0$, since otherwise all its pairings against bounded functions vanish, contradicting its nonzero Hermite action. □

An explicit finite error through degree seven. Here $K \le 3$. Set missing c_k equal to zero and put

$$\lambda_* = \min(\tau, \tau + c_0, \tau + c_1, \tau + c_2, \tau + c_3), \qquad C_* = |\tau - \lambda_*|.$$

Define the following nonnegative rational expressions in d and the absolute coefficient values:

$$\begin{aligned}
B_0 &= |c_0| + |c_1|/2 + 3|c_2|/8 + 5|c_3|/16,\\
B_1 &= \frac{|c_1|/2 + 3|c_2|/4 + 15|c_3|/16}{d+2},\\
B_2 &= \frac{3|c_2|/8 + 15|c_3|/16}{(d+2)(d+4)},\\
B_3 &= \frac{5|c_3|/16}{(d+2)(d+4)(d+6)}.
\end{aligned}$$

Let $J_0 = 1$, $J_j = \prod_{a=1}^{j}(d + 2a - 1)$ and $M_j = \prod_{a=0}^{j-1}(d + 2a)$, with $M_0 = 1$. Put

$$\begin{aligned}
Q_d^* &= C_*^2 + 2C_* \sum_{j=0}^{3} B_j J_j + \frac{1}{d} \sum_{i,j=0}^{3} B_i B_j M_{i+j+1},\\
\mathcal{L}_n(T) &= \frac{d}{n^4}\left[\frac{C_*}{d-2} + \frac{B_0}{d-1} + B_1 + (d+1)B_2 + (d+1)(d+3)B_3\right] + 4\sqrt{Q_d^*}\sqrt{2n\varphi(n)}, \quad d = n^2.
\end{aligned}$$

Then $\mathcal{L}_n(T) = O(n^{-4})$ and

$$V(A_n(T)) \ge q_{n^2}(T) - \mathcal{L}_n(T). \tag{64}$$

In particular, if a separately proved all-sign bound is $N(T) \le B$, the rational reduced sequence satisfies the completely explicit bound

$$G(H_n(T)) \ge \frac{\max(0, q_{n^2}(T) - \mathcal{L}_n(T) - n^{-4})}{B + n^{-4}}.$$

This is a lower bound; it does not identify $N(T)$ with B.

Proof of the error estimate. The radial projection formula gives $TV = t_d(R)V$, where

$$t_d(R) = \tau + \frac{\mathbb{E}R}{d} R \sum_{k=0}^{3} c_k \frac{(1/2)_k}{(d/2+1)_k} L_k^{d/2}(R^2/2).$$

Let v be the clipped-midpoint unit field and $\delta = v - V$. Both fields are odd and unit length, so $2\delta \cdot V = -\|\delta\|^2$. Since the minimum odd eigenvalue is λ_*,

$$\langle v, Tv \rangle \geq q_d(T) - \mathbb{E}[(t_d(R) - \lambda_*)\|\delta\|^2].$$

The coefficientwise absolute Laguerre expansion shows

$$(t_d(R) - \lambda_*)_+ \leq B_d(R) := C_* + \frac{\mathbb{E}R}{d} \sum_{j=0}^{3} B_j R^{2j+1}.$$

On the no-clipping event normalization gives $\|\delta\|^2 \leq dn^{-4}/R^2$; on its complement it is at most four. The latter event has probability at most $2n\varphi(n)$. Gaussian radial moments give

$$\mathbb{E}R^{2j} = M_j, \qquad \mathbb{E}R^{2j+1} = (\mathbb{E}R)J_j, \qquad \mathbb{E}R^{-1} = \frac{\mathbb{E}R}{d-1}, \quad \mathbb{E}R^{-2} = \frac{1}{d-2}.$$

Use $(\mathbb{E}R)^2/d = a_d \leq 1$ in $\mathbb{E}[B_d(R)/R^2]$ and $\mathbb{E}B_d(R)^2$, followed by Cauchy–Schwarz on the clipping event. These are exactly the two terms of $\mathcal{L}_n(T)$. Finally $B_j = O(d^{-j})$, $J_j = O(d^j)$, $M_j = O(d^j)$, which proves the asserted rate. □

Two useful subfamilies. For $T = P_1 - b_3P_3 - b_5P_5 - b_7P_7$ the exact numerator is $a_d - b_3e_3 - b_5e_5 - b_7e_7$. For the normalized identity-tail family

$$T_{\tau,b} = (1-\tau)P_1 - bP_3 + \tau P_{\mathrm{odd}}, \qquad q_d(T_{\tau,b}) = \tau + (1-\tau)a_d - be_3.$$

In the latter family the spectral-cube conditions are $|\tau| \leq 1$ and $|\tau - b| \leq 1$. The identity tail is represented by the diagonal in (62), and is not discarded at finite n. Its numerator has leading loss $(1 - \tau + b)/(2d)$. For every such family, a useful numerical endpoint still requires a proved bound for the complete scalar norm $N(T)$.

Proposition 34 (A powered-dimension cubic sequence)**.** *Fix $0 < b < 1$, and suppose a complete all-sign theorem proves $N(P_1 - bP_3) \leq B$. In the analytic cell-compression construction, take*

$$d_n = n^4, \qquad h_n = n^{-4}, \qquad \textit{coordinate cutoff } n, \qquad \eta_n = n^{-8}, \quad n \geq 2.$$

Here h_n is the interval mesh and η_n is the total entrywise dyadic approximation error. Apply the same exact cloning, symmetric dilation and balanced copy, producing reduced rational matrices $\widehat{H}_n(b)$. Put $d = d_n$ and

$$p_n = \min(1, 2n^3\varphi(n)), \qquad q_d(b) = a_d\left(1 - \frac{b}{2(d+2)}\right),$$

$$\widehat{\mathcal{L}}_n(b) = dh_n^2\left[\frac{1 - b/2}{d-1} + \frac{b}{2(d+2)} + \frac{b}{d-2}\right]$$

$$+ 4\left[1 - \frac{b}{2} + \frac{b}{2}\sqrt{\frac{d+4}{d+2}}\right]\sqrt{p_n} + 4bp_n.$$

Then the explicit finite lower bound is

$$G(\widehat{H}_n(b)) \geq \frac{\max(0, q_d(b) - \widehat{\mathcal{L}}_n(b) - n^{-8})}{B + n^{-8}}.$$

The matrices have actual ratio limit $1/N(P_1 - bP_3)$. The displayed certified lower bounds converge to $1/B$ at rate $O(n^{-4})$. No rate for the actual scalar-norm convergence is asserted.

Proof. For $V = X/R$, the radial projection identity gives

$$(P_1 - bP_3)V = t_d(R)V, \qquad t_d(R) + b = b + \frac{\mathbb{E}R}{d}R\left[1 - \frac{b}{2} + \frac{bR^2}{2(d+2)}\right] \geq 0.$$

The unit-field difference identity and minimum odd eigenvalue $-b$ give a trial-objective loss at most $\mathbb{E}[(t_d(R) + b)\|v - V\|^2]$. Off the clipping event, $\|v - V\|^2 \leq dh_n^2/R^2$. The exact negative radial moments and $a_d \leq 1$ give the first line of $\widehat{\mathcal{L}}_n$. The clipping event has probability at most p_n by the union bound and Gaussian Mills inequality. On that event $\|v - V\|^2 \leq 4$. Cauchy–Schwarz and

$$\mathbb{E}R^2 = d, \qquad \mathbb{E}R^6 = d(d+2)(d+4), \qquad a_d \leq 1$$

give its remaining two terms. These steps use no independence of the clipping event and R.

As before, analytic cell compression has scalar norm at most B and vector norm at least $q_d - \widehat{\mathcal{L}}_n$. Entrywise approximation changes each norm by at most η_n and the exact reductions preserve the ratio, proving the finite bound. The qualitative scalar/vector limits hold for the new nonnested grids because both the dimension and cutoff tend to infinity and the mesh tends to zero. Finally

$$q_{n^4}(b) = 1 - \frac{1+b}{2n^4} + O(n^{-8}), \qquad \widehat{\mathcal{L}}_n(b) = (1+b)n^{-8} + O(n^{-12}) + O(n^{3/2}e^{-n^2/4}).$$

The latter exponential term is $o(n^{-m})$ for every fixed m. Together with $\eta_n = n^{-8}$ this proves the stated lower-bound rate. Qualitative convergence of the scalar norm supplies the actual limit, without supplying a quantitative denominator rate. □

Extension to a normalized odd tail. For $T_{\tau,b} = (1-\tau)P_1 - bP_3 + \tau P_{\text{odd}}$, let $\alpha = 1 - \tau \geq b/2$, and impose the spectral-cube conditions. The same proof holds with $q_d = \tau + \alpha a_d - be_3$ and

$$\widehat{\mathcal{L}}_n(\tau, b) = dh_n^2\left[\frac{\alpha - b/2}{d-1} + \frac{b}{2(d+2)} + \frac{b}{d-2}\right] + 4\left[\alpha - \frac{b}{2} + \frac{b}{2}\sqrt{\frac{d+4}{d+2}}\right]\sqrt{p_n} + 4bp_n.$$

Its leading natural-vector loss is $(\alpha+b)/(2n^4)$, and its midpoint bulk loss is $(\alpha+b)n^{-8}$. A separately certified complete norm bound remains necessary for every selected (τ, b).

Proposition 35 (Exact envelope of criteria with a common quadratic)**.** *Let $P \subset \mathbb{R}^2$ be a bounded convex polygon, allowing a segment, a point, or the empty set. Fix $b > 0$ and finitely many criteria*

$$F_i(u, r) = u^2 + br^2 + A_i u + B_i r + C_i, \qquad 1 \leq i \leq m.$$

For each i, form the closed dominance polygon

$$P_i = P \cap \bigcap_j \{(u, r) : (A_i - A_j)u + (B_i - B_j)r \leq C_j - C_i\}.$$

If P is nonempty, then

$$\max_{(u,r)\in P} \min_i F_i(u,r) = \max_{i:P_i\neq\varnothing} \max_{v\in \mathrm{Vert}(P_i)} F_i(v). \tag{65}$$

For a segment its vertices are its endpoints; for a point the sole vertex is the point. All boundaries and ties are retained. If the inputs are rational, every clipping intersection and every price in (65) *is computed exactly by rational arithmetic.*

Proof. At every point of P at least one affine term $A_iu + B_ir + C_i$ is minimal. Thus the closed P_i cover P; ties simply belong to more than one polygon. On P_i the common quadratic cancels from all comparisons, so $\min_j F_j = F_i$. Each F_i is convex because its Hessian is $\mathrm{diag}(2, 2b) \succ 0$. Expressing any point of P_i as a convex combination of its vertices proves $F_i \le \max_{v\in\mathrm{Vert}(P_i)} F_i(v)$. Taking maxima over the covering polygons proves equality. The same argument handles lower-dimensional polygons, and closed half-plane clipping preserves their boundary points. □

Use in a global all-sign cover. Each C_i must be a proved uniform upper price over the same fixed cumulant–shape cell, and every F_i must bound the same operator at one fixed coefficient b. Then the right side of (65) is an upper bound for every physical sign pair in that source polygon. An arbitrary nonempty subset of the valid criteria is sufficient; it can only increase the pointwise minimum. A floating calculation may nominate this subset, but not replace exact clipping and pricing. An inherited uniform bound may be retained by taking its minimum with this new envelope bound. For rank-one R3 criteria, exact source transport gives the common coefficient b of r^2. More generally a profile is eligible whenever its computed rational r^2 coefficient equals the same b; a profile label is not a substitute for that equality.

H A smaller truncation grid and physical tensor constraints

Proposition 36 (A logarithmic-cutoff reduced sequence)**.** *Fix $0 < b < 1$ and a separately certified complete norm cap $N(P_1 - bP_3) \le B$. For each integer $d \ge 4$, let*

$$R_d = \min\{R \in \mathbb{Z}_{\ge 1} : \exp(R^2/10) \ge d\} = \lceil \sqrt{10 \log d} \rceil.$$

Use mesh $h_d = 1/d$ on $[-R_d, R_d]$ in every coordinate, the two exterior intervals, odd half-cell compression, and total entrywise dyadic approximation error $\eta_d = d^{-2}$. Apply the same exact reduced-class maps. Write the resulting rational matrix as $H_d^*(b)$. Put

$$p_d = \min\left(1, \frac{2d\varphi(R_d)}{R_d}\right), \quad q_d(b) = a_d\left(1 - \frac{b}{2(d+2)}\right),$$

$$\mathcal{L}_d^*(b) = \frac{1}{d}\left[\frac{1-b/2}{d-1} + \frac{b}{2(d+2)} + \frac{b}{d-2}\right] + 4\left[1 - \frac{b}{2} + \frac{b}{2}\sqrt{\frac{d+4}{d+2}}\right]\sqrt{p_d} + 4bp_d.$$

Then

$$G(H_d^*(b)) \ge \frac{\max(0, q_d(b) - \mathcal{L}_d^*(b) - d^{-2})}{B + d^{-2}}.$$

The actual ratios converge to $1/N(P_1 - bP_3)$. The displayed certified lower bounds converge to $1/B$ at rate $O(d^{-1})$, with $\mathcal{L}_d^*(b) = O(d^{-2})$. Each coordinate has only $2dR_d + 2 = O(d\sqrt{\log d})$ intervals, instead of $O(d^{3/2})$ for cutoff $\sqrt{d}$ at the same mesh. The implicit half-cell count is $(2dR_d + 2)^d/2$, before further rational cloning; no practical enumeration guarantee is asserted.

Proof. Mills' inequality and the coordinate union bound give clipping probability at most p_d. From the definition of R_d,

$$\varphi(R_d) \le (2\pi)^{-1/2} d^{-5}, \qquad p_d \le \sqrt{2/\pi}\, d^{-4}/R_d \le d^{-4}.$$

The exact radial/unit-field calculation in the powered-dimension proposition gives $\mathcal{L}_d^*$ after substituting $h = 1/d$ and this clipping probability. Its bulk term is $O(d^{-2})$, and the displayed probability bound gives the same order for its tail. Entrywise approximation and exact cloning/balancing give the finite ratio bound. Both $R_d \to \infty$ and $h_d \to 0$, so conditional averaging of every fixed cylindrical odd function still converges in L_2, proving the qualitative scalar norm limit. The radial unit field proves vector norm convergence to one. Finally $q_d(b) = 1 - (1+b)/(2d) + O(d^{-2})$, proving the stated rate only for the certified lower bounds.

The integer cutoff can be computed without assuming a floating logarithm is exact. Test $\exp(R^2/10) \ge d$ with increasingly precise directed intervals. Equality is impossible for positive R and integer d, because the exponential of a nonzero algebraic number is transcendental. Therefore every comparison terminates and determines the smallest admissible integer. □

Compatibility with a powered index and an odd tail. Taking $d = n^4$ retains certified lower-bound rate $O(n^{-4})$, now with $2n^4 R_{n^4} + 2$ intervals in each coordinate and rational error n^{-8}. For the normalized tail family with $\alpha = 1 - \tau \ge b/2$, replace q_d by $\tau + \alpha a_d - b e_3$ and replace each $1 - b/2$ in the loss formula by $\alpha - b/2$. Its uniform scalar norm needs a separate complete certificate. There is still no quantitative rate claimed for the actual denominator convergence.

Proposition 37 (A necessary cubic cumulant–shape constraint)**.** *Let f be a real symmetric order-three Hilbert tensor of norm one, $Q = I_3(f)/\sqrt{6}$, and*

$$A = \|f \otimes_2 f\|^2, \quad B = \|\operatorname{sym}(f \otimes_1 f)\|^2, \quad \kappa = \mathbb{E}Q^4 - 3, \quad V = \operatorname{Var}(\|DQ\|^2/3).$$

Then

$$\kappa = 36A + 54B, \qquad V = 8A + 6B, \qquad 0 \le B \le A \le 1.$$

For $\kappa > 0$, define the physical shape $z = 9V/\kappa - 1$. It satisfies

$$\frac{2}{5} \le z \le 1, \qquad \kappa z = 36A \le 36. \tag{66}$$

Proof. The product and derivative formulas give

$$Q^2 = 1 + 3I_2(f \otimes_2 f) + \tfrac{3}{2} I_4(\operatorname{sym}(f \otimes_1 f)) + \tfrac{1}{6} I_6(\operatorname{sym}(f \otimes f)),$$

$$\|DQ\|^2/3 = 1 + 2I_2(f \otimes_2 f) + \tfrac{1}{2} I_4(\operatorname{sym}(f \otimes_1 f)).$$

The norm-square of the last order-six tensor is $(2 + 18A)/20$: in the twenty possible groupings of its six indices, two preserve the original grouping and eighteen have contraction inner product A. Chaos orthogonality now proves the two moment identities. Flatten f into a matrix F with one index on its row. Its two Gram matrices have the same nonzero eigenvalues, so $\|f \otimes_1 f\|^2 = \|f \otimes_2 f\|^2 = A$. Orthogonal symmetrization gives $B \le A$. Also FF^T is positive semidefinite with trace one; hence $A = \operatorname{tr}(FF^T)^2 \le 1$. Finally $9V - \kappa = 36A$, and $z = 36A/(36A + 54B)$ gives its stated range. □

Proposition 38 (Complete pricing on the hyperbolic physical boundary)**.** *On a box $\kappa \in [k_-, k_+] \subset [0, 90]$, $z \in [z_-, z_+] \subset [2/5, 1]$, consider*

$$C(\kappa, z) = C_0 + (L_0 + L_1 z)\kappa + D_0 \kappa^2 + \kappa^{3/2} \left[A_0(K - t(z - 2/5)) + s D_0 z^{3/2} \right],$$

where $A_0, D_0 \geq 0$, $K = 194/\sqrt{10}$, $t = 633/9$, $s = 9/4$, and the other coefficients are arbitrary reals. Its maximum on the physical domain $\kappa z \leq 36$ is obtained by the following finite boundary-endpoint check. Put $k_* = \min(k_+, 36/z_-)$ and $k_t = 36/z_+$. If $k_- > k_*$ the domain is empty. Otherwise test

- *(k_-, z_-) and (k_*, z_-);*
- *if $k_- \leq \min(k_*, k_t)$, test (k_-, z_+) and $(\min(k_*, k_t), z_+)$;*
- *if $\max(k_-, k_t) \leq k_*$, test $(\max(k_-, k_t), 36/\max(k_-, k_t))$ and $(k_*, 36/k_*)$.*

Repeated points may be removed. The resulting maximum is exact for this scalar price, including boxes touching the physical boundary. Its point evaluations can be certified by directed arithmetic.

Proof. For fixed κ, the price is convex in z, so its maximum on the feasible vertical interval is at z_- or $\min(z_+, 36/\kappa)$. At fixed z, it is convex in κ: the coefficient of $\kappa^{3/2}$ is nonnegative because $K - t(z - 2/5) \geq 194/\sqrt{10} - 211/5 > 0$. The lower horizontal edge and the upper horizontal portion therefore need only their endpoints. On the remaining hyperbolic edge,

$$C(\kappa, 36/\kappa) = C_0 + 36L_1 + sD_0\, 36^{3/2} + L_0\kappa + D_0\kappa^2 + A_0(K + 2t/5)\kappa^{3/2} - 36A_0 t\sqrt{\kappa}.$$

Each non-affine term here is convex: $D_0, A_0, t \geq 0$ and $-\sqrt{\kappa}$ is convex. Its interval maximum is therefore at the two endpoints as well. The displayed candidates enumerate all these boundary endpoints, proving the claim. The hyperbolic portion starts at $k_t \geq 36$, so no division by zero occurs. □

Application and scope. The frozen R3 moment-price formula has this exact form with $A_0 = q_6 + \zeta^2_{\text{profile}}$ and $D_0 = G - d_0 - \zeta^2_{\text{profile}}$, provided both are nonnegative. The accepted scalar B0 plane prices have the same form with $A_0 = q_6 \geq 0$ and $D_0 = 0$. Thus each can be priced on the necessary physical constraint instead of an unrestricted shape box. Simply deleting infeasible rectangle corners would be invalid; the newly exposed curved-boundary endpoints must also be priced. This lemma does not assert that any resulting numerical cache or full operator cover has already been certified.

Proposition 39 (An optional stronger high-contraction constraint)**.** *In the tensor notation above,*

$$B \geq (3A - 2)_+^2.$$

More strongly, for $A \geq 5/9$,

$$B \geq \left(\frac{3\sqrt{2A - 1} - 1}{2}\right)^2.$$

Writing $w = 36A = \kappa z$, define

$$F(w) = \begin{cases} w, & 0 \leq w \leq 20, \\ \dfrac{31w}{4} - 108 - 81\sqrt{w/18 - 1}, & 20 \leq w \leq 36. \end{cases}$$

Every physical source satisfies $\kappa \geq F(w)$. The function F is increasing and convex; $F(w)/w$ is nondecreasing for $w > 0$. Consequently an explicit stronger shape cap is

$$z \leq 1 \quad (\kappa \leq 20), \qquad z \leq \frac{18}{\kappa}\left[1 + \left(\frac{27 + \sqrt{62\kappa - 1224}}{93}\right)^2\right] \quad (20 \leq \kappa \leq 90).$$

The latter cap decreases from one to $2/5$.

Proof. Let u be a unit top eigenvector of FF^T, with eigenvalue λ. Since its trace is one, $\lambda \geq A$. Symmetry of f means that projection onto u in each tensor leg has squared norm λ. These three projections commute, so the union bound for their complements gives $|f(u,u,u)|^2 \geq 3\lambda - 2$. For $T = \operatorname{sym}(f \otimes_1 f)$,

$$T(u,u,u,u) = \sum_a f(u,u,e_a)^2 \geq |f(u,u,u)|^2, \qquad \|T\|^2 \geq T(u,u,u,u)^2.$$

This proves the first inequality. If $A > 1/2$, the remaining Gram eigenvalues sum to $1-\lambda$, so $A \leq \lambda^2 + (1-\lambda)^2$. Because $\lambda \geq A > 1/2$, this implies $\lambda \geq (1+\sqrt{2A-1})/2$ and proves the stronger inequality when its lower bound is nonnegative.

Substitute it into $\kappa = 36A + 54B$ to obtain $\kappa \geq F(w)$. On its curved part put $s = \sqrt{w/18 - 1} \in [1/3, 1]$. Then $F'(w) = 31/4 - 9/(4s) \geq 1$ and $F''(w) = 1/(16s^3) > 0$; both match the affine part at $w = 20$. Also $wF'(w) - F(w) = 108 + (81/2)(s - 1/s) \geq 0$. The inverse relation is $s = (27 + \sqrt{62\kappa - 1224})/93$, which gives the stated cap. Its monotonicity follows from that of $F(w)/w$, and its endpoint values follow from $F(20) = 20$, $F(36) = 90$. □

Proposition 40 (A stronger physical cubic shape constraint)**.** *Let f be a real symmetric order-three Hilbert tensor with $\|f\| = 1$, and put*

$$Q = I_3(f)/\sqrt{6}, \qquad A = \|f \otimes_2 f\|^2, \qquad B = \|\operatorname{sym}(f \otimes_1 f)\|^2.$$

Then

$$B \geq (2A-1)_+. \tag{67}$$

In particular, if $\kappa = \mathbb{E}Q^4 - 3$ and $z = 9\operatorname{Var}(\|DQ\|^2/3)/\kappa - 1$ for $\kappa > 0$, then

$$\frac{2}{5} \leq z \leq 1, \qquad 0 \leq \kappa \leq 90, \qquad 4\kappa z - \kappa \leq 54. \tag{68}$$

Thus $z \leq \frac{1}{4} + \frac{27}{2\kappa}$ whenever $\kappa > 0$. This last constraint also implies $\kappa z \leq 36$ on $0 \leq \kappa \leq 90$.

Proof. Flatten f to an operator F with one index on its row, and set $G = FF^*$. It is positive with trace one. Its largest eigenvalue λ therefore satisfies $\lambda \geq \operatorname{tr} G^2 = A$. Let u be a corresponding unit eigenvector. If $A > 1/2$, then $\lambda > 1/2$ and

$$A \leq \lambda^2 + (1-\lambda)^2.$$

Indeed, the remaining nonnegative eigenvalues sum to $1-\lambda$, so the sum of their squares is at most $(1-\lambda)^2$. The orthogonal projections onto u in the first and second tensor legs commute. Each projects f to squared norm λ by symmetry, and hence

$$p := \|f(u,u,\cdot)\|^2 \geq 2\lambda - 1 > 0.$$

For the symmetric fourth-order tensor $T = \operatorname{sym}(f \otimes_1 f)$ we have $T(u,u,u,u) = p$. Cauchy–Schwarz and $\|u^{\otimes 4}\| = 1$ give

$$B = \|T\|^2 \geq p^2 \geq (2\lambda - 1)^2 \geq 2A - 1.$$

If $A \leq 1/2$, nonnegativity of B proves the same positive-part bound.

The Gaussian product formulas give $\kappa = 36A + 54B$ and $\operatorname{Var}(\|DQ\|^2/3) = 8A + 6B$, so $\kappa z = 36A$. Also $0 \leq B \leq A \leq 1$, since orthogonal symmetrization is a contraction and the two flattening Gram matrices have the same nonzero eigenvalues. These facts give the first two bounds in (68). If $A > 1/2$, the new contraction bound gives $\kappa \geq 36A + 54(2A-1) = 144A - 54 = 4\kappa z - 54$. If $A \leq 1/2$, then $4\kappa z - 54 \leq \kappa z \leq \kappa$. Finally $(\kappa + 54)/4 \leq 36$ when $\kappa \leq 90$. □

Proposition 41 (Complete pricing on the stronger physical boundary)**.** *Consider a box* $\kappa \in [k_-, k_+] \subset [0, 90]$ *and* $z \in [z_-, z_+] \subset [2/5, 1]$*. Let*

$$C(\kappa, z) = C_0 + (L_0 + L_1 z)\kappa + D_0\kappa^2 + \kappa^{3/2}\left[A_0(K - t(z - 2/5)) + sD_0 z^{3/2}\right],$$

where $A_0, D_0 \geq 0$*,* $K = 194/\sqrt{10}$*,* $t = 633/9$*,* $s = 9/4$*, and* C_0, L_0, L_1 *are arbitrary reals. Its exact maximum on the necessary physical domain* $4\kappa z - \kappa \leq 54$ *is obtained by the following finite endpoint check. Set*

$$c = \tfrac{1}{4}, \qquad d = \tfrac{27}{2}, \qquad k_* = \min(k_+, d/(z_- - c)), \qquad k_t = d/(z_+ - c).$$

The domain is empty if $k_- > k_*$*. Otherwise test*

- (k_-, z_-) *and* (k_*, z_-)*;*
- *if* $k_- \leq \min(k_*, k_t)$*, test* (k_-, z_+) *and* $(\min(k_*, k_t), z_+)$*;*
- *if* $\max(k_-, k_t) \leq k_*$*, test* $(\max(k_-, k_t), c + d/\max(k_-, k_t))$ *and* $(k_*, c + d/k_*)$*.*

Repeated points may be removed. Every candidate is rational for rational box endpoints, and its price may be certified by directed arithmetic.

Proof. For each fixed κ, the price is convex in z, so a maximum occurs at z_- or $\min(z_+, c + d/\kappa)$, with the natural interpretation at $\kappa = 0$. For fixed z it is convex in κ: the coefficient of $\kappa^{3/2}$ is nonnegative because

$$K - t(z - 2/5) \geq 194/\sqrt{10} - 211/5 > 0.$$

The horizontal boundary portions consequently need only their endpoints. On the curved portion, direct substitution gives

$$\begin{aligned} C(\kappa, c + d/\kappa) = {} & C_0 + dL_1 + (L_0 + cL_1)\kappa + D_0\kappa^2 \\ & + A_0[K + t(2/5 - c)]\kappa^{3/2} - A_0 td\sqrt{\kappa} + sD_0(c\kappa + d)^{3/2}. \end{aligned}$$

Each non-affine term is convex in κ: all its displayed coefficients are nonnegative except that of $\sqrt{\kappa}$, whose negative is convex. This portion also needs only its endpoints. The listed candidates are exactly the endpoints of these portions. Since $k_t \geq d/(1 - c) = 18$, the curved portion never divides by zero. The argument retains points and degenerate intervals on the boundary. □

Application and scope. The frozen R3 price has this form when $A_0 = q_6 + \zeta^2_{\text{profile}} \geq 0$ and $D_0 = G - d_0 - \zeta^2_{\text{profile}} \geq 0$. Accepted B0 source-plane prices have $A_0 = q_6 \geq 0$ and $D_0 = 0$. Combined native scalar/L1 endpoint prices also have $D_0 = 0$ and $A_0 = q_6 \geq 0$; their additional shape terms are affine in κ and κz. The same endpoint rule therefore applies to all these guarded families. The physical restriction is an analytic theorem; any numerical improvement still requires fresh directed pricing of every exposed curved endpoint and a complete source cover.

I The first complete refinement

Theorem 42 (The October 2 refined checkpoint)**.** *For the fixed coefficient* $b = 711/1000$*,*

$$\mathcal{N}(b) \leq \frac{140904959}{250000000}, \qquad G_{\mathbb{R}} \geq \frac{250000000}{140904959} = 1.7742455749907283\ldots.$$

The logarithmic-cutoff rational reduced matrices $H_d^*(b)$ satisfy

$$\lim_{d\to\infty} G(H_d^*(b)) = \frac{1}{\mathcal{N}(b)} \geq \frac{250000000}{140904959}.$$

Their explicitly evaluated lower bounds converge at rate $O(d^{-1})$. The actual denominator limit is not replaced by its certificate cap.

Proof. The protected all-dimension source cover and its complete same-b lower branch remain unchanged. The latter covers $0 \leq u \leq 1381/2000$ with price at most $563307766601/10^{12}$. For larger amplitudes every one of the 272579 protected physical source parents is retained, as are the 8737 proved empty cells.

Forty-three complete whole-line local inequalities supplement the old criteria at the single coefficient b. Their exact six spectral gates and all frozen profile/reserve/proof bindings have independent checks. The stronger tensor constraint $4\kappa z - \kappa \leq 54$ is priced using the complete boundary rule above. Independent directed MPFR computations at 768 bits replay 3010 R3 tables with 12040 exposed endpoint prices and 700 selected source planes from 70 source boxes. All underlying 60 source-plane rows are also checked.

The producer-free exact rational geometry replay retains all source polygons and ties, checks 3176 rank envelopes with 7421 closed dominance regions, and checks every knot of the 953 selected native offers. Further subdivision of 4564 amplitude parents yields a closed cover by 19693 children, including 5 proved empty children; all other parents keep their preceding bounds. The child replay checks 994 rank envelopes and 3275 native offers using exactly the same accepted moment prices. Its exact global maximum is

$$\frac{563619834426673}{10^{15}} < \frac{140904959}{250000000}.$$

The smaller-amplitude branch is also below the displayed cap. Thus every physical odd sign pair in every finite Gaussian dimension is covered by the same fixed operator inequality.

The combined independently accepted certificate is `audit/BANK43_STRONG_SHAPEU_REFINED_GLOBAL_ACCEPTANCE.json` in the new-round source tree, with SHA-256

`e7af38399f6e31535645eb9cac7c9ab16b1103b29092bcabc75e294fe16e1663`

It binds the complete geometry, numeric prices, analytic tensor restriction, same-coefficient lower branch, and 593 bound files. The logarithmic-cutoff limit theorem and the exact ratio-preserving reductions prove the matrix-sequence conclusions. □

J Combining source estimates point by point

Proposition 43 (A common affine envelope for source prices)**.** *Let P be a nonempty closed polygon, segment, or point contained in $[u_-, u_+] \times [r_-, r_+]$. Suppose a finite family of independently valid upper bounds for a sign-pair objective has the form*

$$F_i(u, r) = u^2 + \gamma_i r^2 + a_i u + b_i r + c_i.$$

Write $r_0 = (r_- + r_+)/2$ and define the affine functions

$$U_i(u, r) = (u_- + u_+ + a_i)u - u_- u_+ + b_i r + c_i + \begin{cases} \gamma_i((r_- + r_+)r - r_- r_+), & \gamma_i \geq 0, \\ \gamma_i(2r_0 r - r_0^2), & \gamma_i < 0. \end{cases}$$

For each i form the closed dominance region $P_i = P \cap \bigcap_j \{U_i \le U_j\}$, retaining all its vertices even when it is a segment or point. Then the objective is bounded by

$$\max_{i:P_i\neq\varnothing} \max_{v\in\mathrm{vert}(P_i)} U_i(v) = \sup_{(u,r)\in P} \min_i U_i(u,r).$$

Moreover, with $\varepsilon = ((u_+ - u_-)^2 + \max_i |\gamma_i|(r_+ - r_-)^2)/4$,

$$\sup_P \min_i F_i \;\le\; \sup_P \min_i U_i \;\le\; \sup_P \min_i F_i + \varepsilon.$$

All intersections and values are exactly rational when the input is rational. Any nonempty subset of the valid bounds gives a valid upper bound by the same construction.

Proof. For $u \in [u_-, u_+]$, the secant error is $(u_- + u_+)u - u_-u_+ - u^2 = (u - u_-)(u_+ - u)$, between zero and $(u_+ - u_-)^2/4$. For $\gamma_i \ge 0$ the corresponding error in r is $\gamma_i(r - r_-)(r_+ - r)$; for $\gamma_i < 0$ the tangent error is $-\gamma_i(r - r_0)^2$. Thus $F_i \le U_i \le F_i + \varepsilon$ on the rectangle. Taking pointwise minima and then suprema proves both inequalities.

The closed regions P_i cover P, including ties, and on P_i the minimum of the affine functions is U_i. A linear function reaches its maximum on a nonempty compact polygon at a vertex; the same statement holds for a segment or point. This proves the displayed identity. Removing bounds increases their pointwise minimum and therefore preserves validity. □

For a native price that is given by different quadratic formulas between radial knots, first partition the polygon at every such knot and apply the proposition separately to each closed piece. Moment and source constants used on a piece must bound the entire same moment cell; the lemma does not license mixing coefficients from different valid profiles.

K A self-contained upper bound and numerical precision

Proposition 44 (An elementary universal upper bound)**.** *With $c = \log(1 + \sqrt{2})$,*

$$G_{\mathbb{R}} \le \frac{\pi}{2c} < \frac{1760}{987} < \frac{357}{200} = 1.785.$$

Proof. For a real unit vector u, in the Hilbert direct sum of its odd real tensor powers define

$$\Phi(u) = \bigoplus_{k\ge0} \sqrt{\frac{c^{2k+1}}{(2k+1)!}}\, u^{\otimes(2k+1)}, \qquad \Psi(u) = \bigoplus_{k\ge0} (-1)^k \sqrt{\frac{c^{2k+1}}{(2k+1)!}}\, u^{\otimes(2k+1)}.$$

Since $\sinh c = 1$, both vectors have norm one and

$$\langle \Phi(u), \Psi(v)\rangle = \sin\big(c\langle u, v\rangle\big).$$

Also $c < 1$, because $\sinh 1 > 1$, whereas $\pi = 4\int_0^1 (1 + x^2)^{-1}\, dx > 2$; hence $c < \pi/2$. For any finite families u_i, v_j, take a standard Gaussian vector in the finite-dimensional span of their transformed vectors. Put $\epsilon_i = \operatorname{sgn}\langle G, \Phi(u_i)\rangle$ and $\delta_j = \operatorname{sgn}\langle G, \Psi(v_j)\rangle$. Two Gaussian linear forms with correlation ρ have sign-product expectation $2\arcsin(\rho)/\pi$: in their two-dimensional span, rotational invariance makes the angle uniform and their signs differ with probability $\arccos(\rho)/\pi$. Consequently

$$\mathbb{E}[\epsilon_i\delta_j] = \frac{2}{\pi}\arcsin(\sin(c\langle u_i, v_j\rangle)) = \frac{2c}{\pi}\langle u_i, v_j\rangle.$$

For every real matrix A, its scalar sign norm bounds each rounded bilinear form, so

$$\frac{2c}{\pi}\sum_{i,j} a_{ij}\langle u_i, v_j\rangle = \mathbb{E}\sum_{i,j} a_{ij}\epsilon_i\delta_j \le C(A).$$

Taking the vector supremum and then the supremum over matrices proves $G_{\mathbb{R}} \le \pi/(2c)$.

It remains to certify the numerical comparisons using rational arithmetic. The exact identity

$$\frac{x^4(1-x)^4}{1+x^2} = x^6 - 4x^5 + 5x^4 - 4x^2 + 4 - \frac{4}{1+x^2}$$

gives

$$0 < \int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,dx = \frac{22}{7} - \pi.$$

Set $x_0 = 141/160$. The positive exponential series and the successive-tail ratio bound $x_0/(k+1) \le x_0/14$ for $k \ge 13$ give

$$\begin{aligned} e^{x_0} &< \sum_{k=0}^{12} \frac{x_0^k}{k!} + \frac{x_0^{13}}{13!}\frac{1}{1-x_0/14} \\ &= \frac{3654670985963751339938006333048037876 7}{1514001387754963484016640000000000000} < \frac{239}{99} < 1+\sqrt{2}. \end{aligned}$$

The middle comparison follows by clearing positive denominators; the final one follows from $2 - (140/99)^2 = 2/9801 > 0$. Thus $c > 141/160$, and

$$\frac{\pi}{2c} < \frac{22/7}{2(141/160)} = \frac{1760}{987}, \qquad \frac{357}{200} - \frac{1760}{987} = \frac{359}{197400} > 0.$$

All polynomial and rational computations are exactly replayed by `certify_elementary_upper.py`; they require no decimal approximation to π, a logarithm, or a square root. □

Proposition 45 (The three-significant-digit implication)**.** *If a separately accepted complete matrix-sequence certificate proves $G_{\mathbb{R}} \ge 71/40$, then*

$$\frac{71}{40} = 1.775 \le G_{\mathbb{R}} < \frac{1760}{987} < 1.785.$$

Thus $G_{\mathbb{R}}$ rounds to 1.78 to three significant digits. In particular, a complete scalar cap $N(T) \le 40/71$ for a seed whose vector numerator tends to one proves this implication.

Proof. The upper endpoint is Proposition 44. The interval $[1.775, 1.785)$ is precisely the rounding interval for 1.78 at the hundredths place. For the stated seed, its actual matrix ratio limit is $1/N(T) \ge 71/40$, so its matrices give the required lower bound for the unrestricted constant. □

Proposition 46 (Compact evaluation of cubic cell entries)**.** *Let $h_\alpha(x) = \prod_{j=1}^d \mathrm{He}_{\alpha_j}(x_j)/\sqrt{\alpha_j!}$ denote the orthonormal probabilists' Hermite polynomials for standard Gaussian measure. The degree-three projection kernel has the exact formula*

$$\mathcal{K}_3(x,y) := \sum_{|\alpha|=3} h_\alpha(x)h_\alpha(y) = \frac{(x\cdot y)^3 - 3(x\cdot y)(\|x\|^2 + \|y\|^2) + 3(d+2)(x\cdot y)}{6}.$$

For two product cells C, D, let

$$a_{j,k} = \mathbb{E}[X_j^k \mid X \in C], \qquad b_{j,k} = \mathbb{E}[Y_j^k \mid Y \in D], \qquad k = 1, 2, 3,$$

where X, Y are independent standard Gaussian vectors. Put

$$u_j = a_{j,1} b_{j,1}, \quad e_j = (a_{j,2} - 1)(b_{j,2} - 1), \quad t_j = (a_{j,3} - 3a_{j,1})(b_{j,3} - 3b_{j,1}), \quad S = \sum_j u_j.$$

Then

$$\mathbb{E}[\mathcal{K}_3(X, Y) \mid X \in C, Y \in D] = \frac{1}{6} \left\{ \sum_j t_j + 3S \sum_j e_j - 3 \sum_j e_j u_j + S^3 - 3S \sum_j u_j^2 + 2 \sum_j u_j^3 \right\}.$$

Consequently, one compressed cubic matrix entry requires $O(d)$ arithmetic operations after its one-dimensional conditional moments are known. No enumeration of the $\binom{d+2}{3}$ degree-three features is required.

Proof. The degree-three multi-indices have patterns (3), (2, 1) and (1, 1, 1). Since $\mathrm{He}_1(x) = x$, $\mathrm{He}_2(x) = x^2 - 1$ and $\mathrm{He}_3(x) = x^3 - 3x$, their conditional contributions are

$$\frac{1}{6} \sum_j t_j + \frac{1}{2} \sum_{i \neq j} e_i u_j + \sum_{i<j<k} u_i u_j u_k.$$

Use $\sum_{i \neq j} e_i u_j = S \sum_i e_i - \sum_i e_i u_i$ and $6 \sum_{i<j<k} u_i u_j u_k = S^3 - 3S \sum_i u_i^2 + 2 \sum_i u_i^3$. Product conditioning preserves the independence of the coordinates, so these identities prove the stated expectation. Substituting the point moments $a_{j,k} = x_j^k$, $b_{j,k} = y_j^k$ and collecting terms gives the displayed projection kernel.

For an interval $I = (\ell, r)$ with positive Gaussian mass $p_I = \Phi(r) - \Phi(\ell)$, its required conditional moments are

$$m_1(I) = \frac{\varphi(\ell) - \varphi(r)}{p_I}, \qquad m_2(I) = 1 + \frac{\ell\varphi(\ell) - r\varphi(r)}{p_I},$$
$$m_3(I) = \frac{(\ell^2 + 2)\varphi(\ell) - (r^2 + 2)\varphi(r)}{p_I}.$$

The expressions at infinite endpoints are interpreted by their zero limits. These formulas follow by integrating $\varphi'(x) = -x\varphi(x)$.

For a selected half-cell C in the antipodal fold, let $w_C = 2\gamma_d(C)$. The analytic cubic seed entry is exactly

$$A_{C,D} = w_C w_D \left(S - b\, \mathbb{E}[\mathcal{K}_3(X, Y) \mid C, D] \right).$$

Indeed, every degree-one or degree-three Hermite cell feature equals $w_C \mathbb{E}[h_\alpha(X) \mid C]$, so its outer product is exactly the folded cell compression. Rational approximation and the reduced matrix maps are applied subsequently as before; this entry formula changes neither the sequence nor its certified bounds. □

Corollary 47 (A fully rational finite bound)**.** *In Proposition 36, assume $0 < b < 1$, $d \geq 4$ and $N(P_1 - bP_3) \leq B$. Define the rational numbers*

$$\underline{q}_d = \left(1 - \frac{1}{2d}\right) \left(1 - \frac{b}{2(d+2)}\right),$$

$$\overline{L}_d = \frac{1}{d}\left(\frac{1-b/2}{d-1} + \frac{b}{2(d+2)} + \frac{b}{d-2}\right) + \frac{4}{d^2}\left(1 + \frac{b}{2(d+2)}\right) + \frac{4b}{d^4}.$$

Then the same rational reduced matrices satisfy

$$G(H_d^*(b)) \geq \frac{\max(0, \underline{q}_d - \overline{L}_d - d^{-2})}{B + d^{-2}}.$$

If B is rational, this bound uses only rational arithmetic. It does not require a numerical gamma-function evaluation or a Gaussian tail evaluation. The right-hand side converges to $1/B$ at rate $O(d^{-1})$.

Proof. For $R = \|X\|$, its centered Hermite expansion has only even degrees, starting at degree two. The Hermite derivative identity therefore gives

$$\mathrm{Var}(R) \leq \frac{1}{2}\sum_{k\geq 1} k\|P_k R\|_2^2 = \frac{1}{2}\mathbb{E}\|\nabla R\|^2 = \frac{1}{2}.$$

Consequently $a_d = (\mathbb{E}R)^2/d \geq 1 - 1/(2d)$, giving $q_d(b) \geq \underline{q}_d$. The logarithmic cutoff gives $p_d \leq d^{-4}$. Finally,

$$\sqrt{\frac{d+4}{d+2}} = \sqrt{1 + \frac{2}{d+2}} \leq 1 + \frac{1}{d+2}.$$

Substitute these inequalities into the stated midpoint-error bound, using $\sqrt{p_d} \leq d^{-2}$ and the positive coefficients of its tail terms. Thus $\mathcal{L}_d^*(b) \leq \overline{L}_d$. The finite ratio bound of the logarithmic-cutoff proposition now proves the assertion. Since $\underline{q}_d = 1 - (1+b)/(2d) + O(d^{-2})$ and $\overline{L}_d = O(d^{-2})$, the stated rate follows. □

L Final constructive enclosure and finite matrices

Theorem 48 (Certified constructive enclosure)**.** *Let $b = 711/1000$ and $B_* = \frac{40}{71}$. Then*

$$\mathcal{N}(b) := N(P_1 - bP_3) \leq B_*, \qquad G_{\mathbb{R}} \geq \frac{71}{40} = 1.775.$$

For the logarithmic-cutoff rational reduced matrices of Proposition 36,

$$\lim_{d\to\infty} G(H_d^*(b)) = \frac{1}{\mathcal{N}(b)} \geq \frac{1}{B_*}.$$

In particular the matrices are symmetric, have zero diagonal and zero row sums, and their entrywise absolute values are doubly stochastic. For every stated finite index their certified lower bound is

$$G(H_d^*(b)) \geq \frac{\max(0, q_d(b) - \mathcal{L}_d^*(b) - d^{-2})}{B_* + d^{-2}}.$$

These lower formulas converge to $1/B_*$ at rate $O(d^{-1})$, or $O(n^{-4})$ on $d = n^4$. This rate does not assert a quantitative rate for the actual scalar denominator. Together with Proposition 44,

$$1.775 \leq G_{\mathbb{R}} < \frac{1760}{987} < 1.785.$$

Thus $G_{\mathbb{R}}$ rounds to 1.78 to three significant digits.

Proof. The complete same-coefficient lower-amplitude proof covers $0 \le u \le 1381/2000$ with cap $563307766601/10^{12}$. The accepted upper cover retains all 272579 physical source parents, 8737 inherited proved-empty cells, and 19693 closed amplitude children. The shared cumulant–shape refinement has 18480 inherited closed product children. The final step replaces exactly two of them by their eight closed Cartesian descendants, giving 18486 current closed moment children. Every inherited bound remains available; any empty child is removed only by an accepted necessary physical constraint or a uniformly valid source halfspace.

There are 89 independently accepted complete whole-line criteria. The accepted affine parent has 5280 independently repriced uniform moment tables in addition to the complete shared-moment ancestor. The final four moment boxes have 356 uniform tables: 344 are checked equal to previously accepted tables and the remaining 12 are freshly repriced with directed arithmetic. The four corresponding source-plane boxes are also independently repriced. These constants cover the actual shared physical moment domain, and every spectral gate and literal remainder allowance is bound to the corresponding complete local proof. Such criteria remain valid outside the box in which they were discovered. The full even Gram criteria retain their actual radial quadratic coefficient, including negative coefficients.

At each source cell all radial knots are split before a native chord is used, and each chord retains one full native kernel. Uniform quadratic prices are replaced by valid affine secants or tangents. The closed dominance polygons of Proposition 43 cover all ties and degenerate source sets. Their exact vertex maxima, with every old cap retained, prove the displayed complete upper bound. The final eight descendants are all nonempty and pass the exact threshold $40/71$; there is no uncovered or failing source cell. Any further closed source subdivision is checked against its parent and retains the entire parent domain. The lower and upper amplitude proofs use the same coefficient b and cover $[0, \sqrt{2/\pi}]$.

The complete independent aggregate is `FINAL_GOAL_AFFINE_MOMENT_GLOBAL_ACCEPTANCE.json`. It binds every local, numerical, geometric, ancestor and lower-amplitude input. The final envelope has SHA-256

`bdcde16a1efcc9a1e414236394ea0edb1f63722e3711bcd1a01c2530e005c8ff`

The Gaussian-to-matrix limit and exact reduced-class maps prove the constructive matrix claims. The unrestricted upper corollary gives the rounding interval. Neither the scalar cap nor this matrix-family limit is identified with the exact unrestricted Grothendieck constant. □

Finite matrices evaluated with rational arithmetic. Taking $B = 40/71$ in Corollary 47 gives the following certified lower bounds. Every displayed decimal is rounded down to twelve places from an exact rational value:

Gaussian dimension d	Lower bound for $G(H_d^*(711/1000))$
10^3	1.773467997012
10^4	1.774848013893
10^6	1.774998481474
10^8	1.774999984814

The index d is the Gaussian dimension. The matrix order includes the product-cell compression and the subsequent exact cloning maps; these large matrices are specified implicitly. All total entrywise rational approximation errors d^{-2} are charged in the displayed bounds. The exact arithmetic inputs are recorded in `FINAL_GOAL_RATIONAL_FINITE_BOUND.json`. These finite lower certificates converge to $71/40$. The actual matrix ratios converge to $1/\mathcal{N}(711/1000)$, which is at least $71/40$.

The limiting inequality and the unrestricted upper bound together prove the three-significant-digit conclusion.

The source package also supplies independent Arb768 and MPFR768 evaluations of the exact logarithmic-cutoff formula on $d = n^4$, joined to the same accepted norm proof by `FINAL_GOAL_FINITE_FORMULA_ACCEPTANCE.json`. No numerical extrapolation is used to prove the limit.

Replaying the completed proof. The source package retains every explicitly bound input and the complete historical source inventory. The frozen current input whitelist is `FINAL_GOAL_INPUT_MANIFEST.json`; its replay guide is `README_GOAL_REPLAY.md`. The `--all-local` mode reconstructs the protected baseline, every current whole-line criterion, the directed moment and source prices, the exact closed partition and dominance geometry, and the elementary upper-bound arithmetic. Every criterion uses its own literal remainder allowances and its certified tail cutoff. All input hashes are checked before and after replay; accepted certificate bytes are preserved and new outputs are written separately.